

A Survey of Talking Head Synthesis Techniques: Portrait Generation, Driving Mechanisms, and Editing

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Talking head synthesis, an advanced method for generating portrait videos from a still image driven by specific content, has garnered widespread attention in virtual reality, augmented reality and game production. Recently, significant breakthroughs have been made with the introduction of novel models such as the transformer and the diffusion model. Current methods can not only generate new content but also edit the generated material. This survey systematically reviews the technology, categorizing it into three pivotal domains: portrait generation, driving mechanisms, and editing techniques. We summarize milestone studies and critically analyze their innovations and shortcomings within each domain. Additionally, we organize an extensive collection of datasets and provide a thorough performance analysis of current methodologies based on various

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evaluation metrics, aiming to furnish a clear framework and robust data support for future research. Finally, we explore application scenarios of talking head synthesis, illustrate them with specific cases, and examine potential future directions. Moreover, we present a first-of-its-kind system-level analysis of integration-specific challenges—such as identity consistency drift, interface incompatibility, and real-time synchronization—that emerge only when portrait generation, driving, and editing are jointly deployed.

CCS Concepts: • **Computing methodologies** → **Computer vision**;

Additional Key Words and Phrases: Talking head synthesis, portrait generation, avatar animation, facial editing

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1 Introduction

The rapid advancements in deep learning have catalyzed significant progress in the field of computer vision. One of the landmark innovations in this domain is the development of **Generative Adversarial Networks (GANs)** [56], which has dramatically transformed image generation and manipulation tasks. This technology extends beyond simple image editing techniques, such as style transfer, image inpainting, and super-resolution, to the sophisticated generation of entirely new images from textual descriptions or random noise. Among these generative tasks, one of the most compelling challenges is the creation of realistic and natural human portraits that avoid the uncanny valley effect. Talking head synthesis, a specialized application focused on generating dynamic, speaking portraits from static images, has demonstrated significant potential across various domains. Not only does it foster the development of the digital economy, reduce video production costs, and enhance content diversity, but it also significantly improves the efficiency and immersion of remote communication through real-time portrait driving and cross-language automatic lip synchronization. Furthermore, virtual teachers and doctors help reduce disparities in educational and healthcare resources, virtual characters also play a crucial role in providing psychological support and emotional companionship. However, existing surveys tend to concentrate on specific subfields or singular driving approaches, lacking a comprehensive technological narrative that spans from static images to dynamic videos. To address this gap, our review systematically summarizes and evaluates the core methods, key challenges, and future prospects of talking head synthesis.

Talking head synthesis encompasses three primary stages: portrait generation, driving mechanisms, and editing techniques. The initial stage, portrait generation, involves creating new facial images using advanced generative models such as GANs, **Variational Autoencoders (VAEs)** [90], and Diffusion Models [68]. These models enable the production of high-quality static images that serve as the foundation for subsequent animation. The second stage, driving mechanisms, is critical for animating these static images. This process can be driven by video or audio inputs, which dictate the movements and expressions of the generated portraits. Video-driven approaches replicate the facial movements and expressions of a character from a driving video [167, 182], while audio-driven methods synchronize lip movements and facial expressions with an audio track [98, 211]. Text inputs, often converted to audio before driving, also play a significant role in this stage. Key factors in this process include preserving the identity of the source image, accurately replicating head movements, and capturing the nuances of facial expressions. The final stage, editing techniques, allows for the modification of generated portraits to enhance realism and suitability for

specific applications. These techniques enable the adjustment of attributes such as emotions, head orientation, and eye blinking without necessitating retraining of the model [28, 92, 160, 165]. This flexibility is crucial for refining the results and achieving high-quality outputs. A generated image or a captured portrait can be driven to produce the corresponding animation, with further detailed modifications facilitated through editing techniques. The collaboration of these three stages allows for the generation of precise, high-quality facial animations.

Despite the remarkable advancements in talking head synthesis, several challenges persist. These include maintaining temporal consistency, handling large-angle head poses, and addressing data biases. Furthermore, the reliance on large pre-trained models poses limitations in terms of adaptability and computational requirements. Addressing these challenges is essential for advancing the field and expanding its applications. This review centers on the field of talking head synthesis, we delve it into three pivotal domains of portrait generation, driving mechanisms, and editing techniques, following the order of animation generation from static to dynamic. When selecting representative papers, we prioritize influential journals and conferences in the field of computer vision or highly-cited works, ensuring the professionalism and authority of the review. We selected literature from 2019 to 2024, covering the latest technological advancements in talking head synthesis and introducing milestone representative studies. In this comprehensive survey, we systematically highlight key studies and critically analyze their innovations and limitations. Additionally, we compile and organize extensive datasets, providing a thorough performance analysis based on various evaluation metrics. We also explore the diverse application scenarios of talking head synthesis, presenting specific case studies and discussing future research directions. This survey aims to provide a clear framework and robust data support for researchers and practitioners, facilitating further advancements in this rapidly evolving field.

1.1 Comparison with Existing Surveys

In this section, we provide a comparative analysis of our work with recent surveys on talking head synthesis.

Kammoun et al. [83] reviewed the application of GANs in face generation, focusing on direct face synthesis using GANs. In contrast, our survey delves into the utilization of audio and video inputs for facial animation, offering a broader perspective.

Zhang et al. [219] examined text-guided image generation with diffusion models, highlighting the advancements in generating images from textual descriptions. Our work, however, centers on the dynamic aspect of facial animation driven by audio and video, presenting a different angle of the generative model applications.

Liu et al. [109] provided a comprehensive overview of audio-driven talking head synthesis. While our paper acknowledges the contributions in audio-driven methods, it expands the scope to include video-driven techniques, offering a more holistic view of the field.

Sha et al. [149] and Gowda et al. [57] contributed extensive reviews on talking head synthesis. Our paper advances the discourse by incorporating a detailed examination of editing techniques and the latest methodologies for integrating diffusion models into talking head synthesis.

Our review offers a systematic exploration of the three pivotal components of talking head synthesis: portrait generation, driving mechanisms, and editing techniques. Figure 1 illustrates the chronological development of the three primary stages of talking head synthesis. We meticulously analyze the application of avant-garde technologies, such as GANs and diffusion models, to assess their current capabilities and delineate potential applications. Moreover, we underscore the existing challenges and chart a course for future research directions.

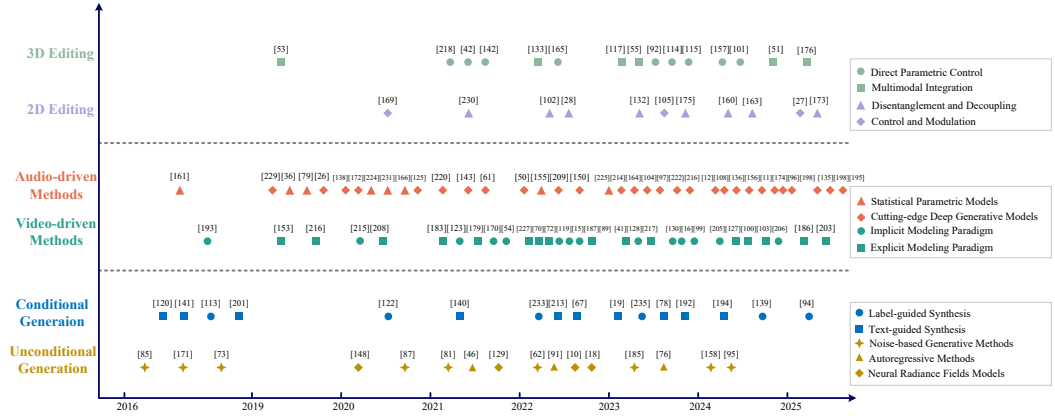


Fig. 1. Chronological overview of the talking head synthesis's three primary stages. The bottom-up order is portrait generation, driving mechanisms and editing techniques. Pentagrams mark more influential methods in each stage.

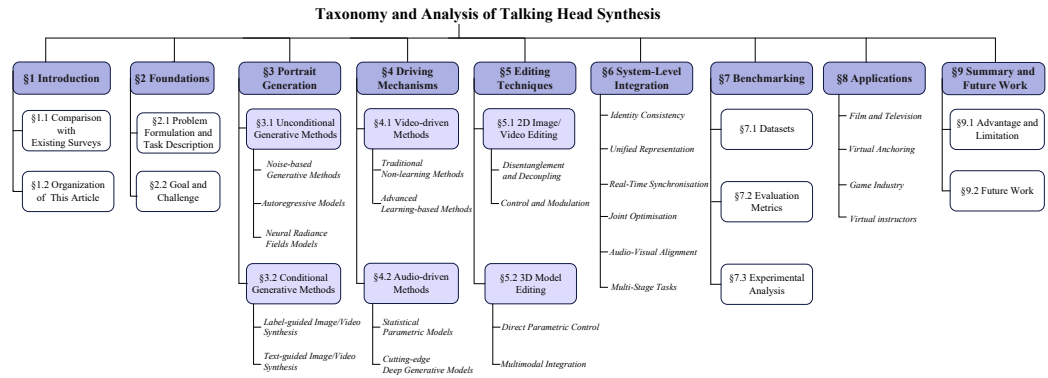


Fig. 2. An overview of this survey.

1.2 Organization of This Article

Our paper presents a comprehensive survey of talking head synthesis. Section 2 introduces the foundations, covering fundamental algorithmic processes and task descriptions (Section 2.1) and summarizing existing problems and challenges (Section 2.2). Section 3 explores generative methods, detailing unconditional methods (Section 3.1) and conditional methods (Section 3.2). Section 4 describes animation driving methods, including video-driven (Section 4.1) and audio-driven techniques (Section 4.2). Section 5 reviews editing techniques, divided into 2D Image/Video Editing (Section 5.1) and 3D Model Editing (Section 5.2). Section 6 analyzes vital challenges of system-level integration in end-to-end pipeline of talking head synthesis. Section 7 presents benchmarks, including datasets, evaluation metrics, and experimental analysis. Section 8 discusses applications, while Section 9 addresses the advantages, limitations, existing issues, and potential solutions for future work. Finally, Section 10 concludes the survey with conclusive remarks, encapsulating the collective findings and underscoring the significance of the research in the domain of talking head synthesis. This organization is visually represented in Figure 2.

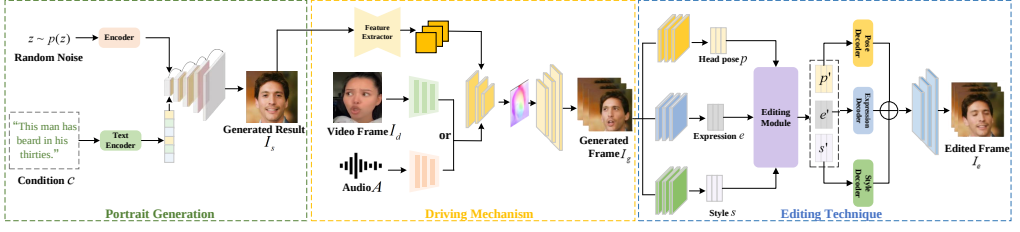


Fig. 3. The comprehensive process of talking head synthesis involves three key stages. First, portrait generation creates a static image from random noise, optionally incorporating specified attributes. Second, the driving mechanism animates this image using video frames or audio content through the extraction and application of intermediate representations like keypoints and landmarks. Third, the editing technique refines the animated output, enhancing visual coherence and quality by precisely adjusting pose and expression parameters.

2 Foundations

2.1 Problem Formulation and Task Description

Talking head synthesis is a multifaceted task designed to generate, animate, and refine realistic human head images based on specific conditions and driving content. This process is crucial for applications such as virtual avatars, video conferencing, and entertainment. The pipeline can be divided into three primary stages: talking head portrait generation, talking head driving, and talking head editing. Each stage plays an essential role in ensuring that the final output is both visually coherent and dynamically expressive. The entire process of talking head synthesis is illustrated in Figure 3. In this context, we use a generative model to create a portrait of a person from random noise $z \sim p(z)$. When specific results are desired, additional conditions can be included as inputs to the model. The image generation process can be described as follows:

$$I_s = G_I(z, c; \beta_{G_I}), \quad (1)$$

where I_s is the generated image, c is the condition as a guidance for generation, G_I is the generative model and β_{G_I} means model's parameters.

In the process of talking head driving, there are mainly two driven contents: video and audio. For video-driven methods, both a source image and a driving video V_d are required. We select the generated image I_s as the source image, although any portrait image can be used. The frames in the video are denoted as I_d . Using I_s and I_d , we aim to generate a new image I_g . The generation process can be divided into four detailed steps: intermediate representation, feature extraction, transformation, and image generation. Firstly, the source image I_s and the video frame I_d are processed by a network F , which predicts or detects intermediate representations, such as keypoints, landmarks, and 3D meshes. This can be formulated as:

$$R_s = F(I_s; \beta_F), R_d = F(I_d; \beta_F), \quad (2)$$

where R_s and R_d are the intermediate representations of the source image and video frame, respectively, β_F is the parameters of the network F . The intermediate representations will be used to transform the intermediate representation of the source image, $\bar{R}_s$ preserves the identity of the source image but has same pose and expression with the video frame. This can be described as:

$$\bar{R}_s = T(R_s, R_d; \beta_T), \quad (3)$$

where T is the transformation network and β_T is the parameters of it. Finally, we need convert the intermediate representation $\bar{R}_s$ to the image I_g and obtain the generated V_d . This can be

expressed as:

$$I_g = G(\bar{R}_s; \beta_G), \quad (4)$$

where G is the generative network and β_G is the parameters of it. For audio-driven talking head synthesis, the input is a source image I_s and a audio clip A used for driving. The processing of the source image is the same as in video-driven methods, first obtaining an intermediate representation, then it is deformed under the influence of features extracted from the audio. However, the processing of audio is more complex as audio and image belong to different modalities of information. It is necessary to firstly extract features from the audio, such as the Mel Spectrogram, Fundamental Frequency (F0), and vocal intensity. Afterwards, a mapping model is needed to convert the extracted audio features into control signals for facial movements. These steps can be represented as:

$$\bar{R}_s = T(R_s, C(M(A))), \quad (5)$$

where M is an audio feature extractor and C is a mapping network. Then, $\bar{R}_s$ will be the input to the generative network to get I_g . Overall, audio-driven talking head synthesis focuses on utilizing audio features to generate accurate mouth movements and corresponding facial expressions, while video-driven methods emphasize capturing more comprehensive facial movements and expressions from video.

For the task of editing, we can understand it as a higher-level operation. Not only can it accomplish driving, but it also adds an editing module. The module enables us to modify specific attributes during talking head synthesis. Typically, editing requires the control of decoupled parameters to generate an newly edited video V_e , such as the parameters of **3D morphable model (3DMM)** [13]. This task can be clearly described as:

$$I_e = E(I_g; \beta_E), \quad (6)$$

where E is the editing module, I_e is the edited image and β_E is the parameters of it.

2.2 Goals and Challenges

Talking-head synthesis seeks to produce photorealistic, temporally coherent videos of a specific individual speaking, guided by audio, text, or motion signals. This task uniquely demands unwavering identity fidelity, rich expressive dynamics—including accurate lip motion, head pose, and micro-expressions—precise audio-visual alignment at sub-frame granularity, and low-latency responsiveness that supports on-the-fly personalization.

Each functional stage presents formidable hurdles: portrait generation suffers from a scarcity of large, diverse, high-resolution datasets needed to capture identity variations across lighting and demographics; driving mechanisms must generalize from controlled settings to extreme head rotations and subtle affective cues; editing techniques require fine-grained, attribute-specific control without compromising overall coherence. When these stages are composed into a unified pipeline, additional system-level difficulties emerge. Ensuring identity consistency across latent embeddings, key-point or mesh deformations, and parametric edits is critical to prevent cumulative drift. The lack of a standardized intermediate representation forces bespoke translators at every interface, complicating joint optimization. Disparities in computational load further introduce latency asymmetries that can disrupt seamless real-time interaction. Moreover, stage-wise training under separate objectives precludes holistic optimization; devising an end-to-end differentiable framework that reconciles identity preservation, expressive fidelity, temporal continuity, and efficiency remains an open challenge.

Compared to related multi-stage endeavors—such as full-body motion synthesis or object-centric video generation—talking head synthesis enforces stricter perceptual thresholds on lip-speech synchrony, micro-expression realism, and identity preservation, while also demanding interactive,

personalized control. These compounded requirements magnify the integration challenges above. Section 6 provides a comprehensive analysis of these system-level issues, and Section 9.2 outlines promising directions to address both stage-specific and integration-level obstacles.

3 Portrait Generation

Portrait generation focuses on the automated creation of digital images depicting human faces, primarily utilizing generative models, with GANs being the most prominent. Research in this field is divided into two main categories based on the nature of the input content: unconditional generative methods and conditional generative methods. Table 1 summarizes the representative works on portrait generation.

3.1 Unconditional Generative Methods

Unconditional generative methods create digital portraits without relying on specific input conditions such as attributes, labels, or guiding text. These methods leverage random noise or inherent model structures and can be grouped into three main categories: noise-based generative methods, which utilize random noise; **autoregressive models (ARG)**, which model probability distributions; and NeRF models, which represent scenes as continuous volumetric functions. The term “scene” in this context specifically refers to the human head, as originally defined broadly in NeRF literature but specifically tailored to our study’s focus.

3.1.1 Noise-based Generative Methods. Noise-based methods utilize random noise as input and map it to the data space through neural network models to generate new data samples. These methods learn the mapping relationship between noise and data to produce realistic outputs.

Recent advancements in generative models have significantly improved the quality and efficiency of both image and video synthesis. Karras et al. [85] pioneered enhancements in image generation by refining generator architecture and training methods, including redesigned normalization, revised progressive growing, and path length regularization. In the realm of video generation, MoCoGAN [171] innovatively separated motion from content by merging fixed content vectors with stochastic motion vectors, enabling dynamic video generation. Its adversarial learning framework incorporated both image and video discriminators for effective content-motion decomposition. IntroVAE [73] combined the stability of VAEs with the adversarial nature of GANs, training the generator and inference models introspectively to produce high-resolution videos through self-evaluation and refinement. Building on the improvements of [85], Karras et al. [87] further refined the approach by incrementally growing the generator and discriminator, accelerating training and supporting high-resolution outputs. Jiang et al. [81] addressed the challenge of training GANs with limited datasets by introducing **Adaptive Pseudo Augmentation (APA)**, which leverages the generator to expand the real data distribution, preventing discriminator overfitting and enhancing image synthesis quality in data-scarce environments. RV-GAN [62] introduced the **TransConv LSTM (TC-LSTM)**, a novel recurrent GAN architecture that captures spatial and temporal relationships within a single LSTM unit, enabling superior video synthesis. Wang et al. [185] designed a new motion generator based on a learning-based inversion network, enhancing video generation by constraining the generator’s latent space to incorporate smooth priors and leveraging a pre-trained StyleGAN image generator for high-quality video generation with style transfer capabilities.

Noise-based generative methods, including GANs and VAEs, are highly flexible in modeling complex data distributions and learning meaningful latent representations, enabling the generation of high-fidelity samples across various domains [107]. However, they face challenges such as mode collapse, training instability, and evaluation difficulties, which can restrict diversity and require

Table 1. The Characteristics of Representative Portrait Generation Works

Method	Year	Venue	Models	Outputs.R	Outputs.T	Controllable
Unconditional Generative Methods						
PG-GAN [85]	2018	ICLR	GAN	1,024×1,024	Image	
MoCoGAN [171]	2018	CVPR	GAN	N/A	Video	✓
IntroVAE [73]	2018	NeurIPS	VAE	1,024×1,024	Video	
StyleGAN [87]	2020	CVPR	GAN	N/A	Image	✓
GRAF [148]	2020	NeurIPS	GAN + NeRF	512×512	Image	✓
ImageBART [46]	2021	NeurIPS	Autoregressive	256×256	Image	✓
Deceive D [81]	2021	NeurIPS	GAN	256×256	Image	
GIRAFFE [129]	2021	CVPR	GAN + NeRF	256×256	Image	✓
RQ-AIG [91]	2022	CVPR	Autoregressive+VAE	256×256	Image	
3D-GNeRF [10]	2022	NeurIPS	GAN + NeRF	N/A	Image	
RV-GAN [62]	2022	CVPR	GAN	N/A	Video	
StyleInV [185]	2023	ICCV	Diffusion	256×256	Image	
TAIC [76]	2023	CPVR	Autoregressive	N/A	Image	
AlbedoGAN [158]	2024	WACV	GAN	N/A	Image	✓
UV-IDM [95]	2024	CVPR	Diffusion	512×512	Image	
Conditional Generative Methods						
AlignDRAW [120]	2016	ICLR	RNN	N/A	Image	✓
T2I-GAN [141]	2016	ICML	GAN	N/A	Image	✓
AG-CGAN [113]	2018	ECCV	GAN	256×256	Image	✓
AttnGAN [201]	2018	CVPR	GAN	256×256	Image	
AD-GAN [122]	2020	CVPR	GAN	256×256	Image	✓
DALL-E [140]	2021	ICML	Autoregressive	N/A	Image	✓
Parti [213]	2022	TMLR	Autoregressive	1,024×1,024	Image	✓
Composer [75]	2023	ICML	Diffusion	1,024×1,024	Image	✓
4D-EDM [235]	2023	TOMM	Diffusion	N/A	Image	✓
TB-GAN [19]	2023	WACV	GAN	N/A	Image	✓
Tune-A-Video [192]	2023	ICCV	Diffusion	N/A	Video	✓
Collaborative-DM [78]	2023	CVPR	Diffusion	512×512	Image	✓
Portrait3D [194]	2024	ACM	GAN	N/A	Image	
CemiFace [139]	2024	NeurIPS	GAN	N/A	Image	✓
StyO [94]	2025	AAAI	Diffusion	512×512	Image	✓

‘Year’ refers to the publication year. ‘Venue’ indicates the conference or journal where the study was published. ‘Models’ indicates the types of models used. ‘Outputs.R’ refers to the resolution of the generated output. ‘Outputs.T’ describes the type of the output, such as images or videos. ‘Controllable’ means the ability to control certain attributes during the generation process, enhancing customization.

meticulous hyperparameter tuning [14]. To overcome these issues, ARG have been developed, leveraging probabilistic dependencies to produce coherent and diverse outputs.

3.1.2 Autoregressive Models. ARG formulate the generation task as a conditional probability distribution, progressively generating each element in a sequence based on the previously generated elements, thereby producing a complete data sample. Esser et al. [46] introduced ImageBART, an innovative ARG model that integrates a multinomial diffusion process and employs a coarse-to-fine context hierarchy to enhance image synthesis. This model excels in high-fidelity image generation and modification, benefiting from efficient training in a compressed latent space, and supports both unconditional and conditional image generation with user-driven local editing. To overcome the limitations of traditional ARG models, such as missing broader scene information, Lee et al. [91] proposed a two-stage framework comprising the **Residual-Quantized Variational Autoencoder (RQ-VAE)** to compress high-resolution images into discrete codes and the RQ-Transformer to model sequence dependencies, enabling rapid generation of high-fidelity images. Huang et al. [76] further advanced the field by introducing **DynamicQuantization VAE (DQ-VAE)** and DQ-Transformer,

which encode image regions into variable-length codes based on information density, employing a novel transformer architecture to improve the quality and speed of image generation.

The above methods of ARG excel in capturing sequential dependencies and long-range correlations, generating coherent, and realistic samples in portrait generation. However, ARG typically require a large amount of training data to ensure high-quality outputs in high-fidelity portrait generation. Additionally, these models rely on a linear structure where each element is predicted based on its predecessor, which limits their ability to capture long-range spatial and temporal dependencies—especially in high-complexity 3D scenes [47]. In contrast, **Neural Radiance Field (NeRF)** models require comparatively fewer training samples and perform exceptionally well in person-specific applications, thereby significantly reducing the data collection burden. NeRF can render high-quality 3D images from arbitrary viewpoints, making it particularly effective for generating visually complex and detail-rich scenes. This capability is especially critical for high-fidelity portrait generation, where capturing fine details is paramount.

3.1.3 Neural Radiance Fields Models. NeRF are capable of rendering high-quality 3D scene images from any perspective [9, 189]. In portrait generation, the process begins by producing a NeRF through generative models, which then renders new images by altering the viewpoint to capture previously invisible parts. One of the first attempts at this was GRAF [148], which proposed a generative model for radiance fields to achieve precise control over camera viewpoint or object pose. Given a random code and a view direction as NeRF's input, GRAF can synthesize high-resolution images. Niemeyer et al. [129] advanced this by adopting compositional generative neural feature fields, effectively disentangling individual objects, shapes, and appearances from the background for better manipulation. Bergman et al. [10] introduced the first radiance field for human bodies and facial expressions. They developed a 3D GAN to generate radiance fields representing human bodies and faces in a canonical pose, then warped them through a deformation field. Pix2NeRF [18] was the first work on NeRF-based GAN inversion, leveraging π -GAN, an unconditional generative model, to transfer random latent codes into object-specific radiance fields.

NeRF enhance traditional generative methods by providing detailed and spatially coherent 3D reconstructions, making them particularly effective for generating images from new viewpoints. However, NeRF-based methods are computationally demanding and exhibit poor generalization to novel scenarios, rendering them unsuitable for real-time applications and necessitating meticulous tuning for optimal performance [40]. In unconditional generative methods, the model is tasked solely with learning the intrinsic distribution of the data without relying on additional labels or conditional inputs. While this approach simplifies data preparation and model design, it lacks the flexibility to specify precise attributes of the generated samples, thus limiting control over the output content. To address the shortcoming in controllability, researchers have developed conditional generative methods that incorporate explicit conditions to guide the generation process, enhancing both control and efficiency.

3.2 Conditional Generative Methods

Conditional generative methods enhance unconditional techniques by incorporating specific conditions or parameters to guide image synthesis. It enables the generation of images with predefined attributes, such as labels or textual prompts, allowing precise feature manipulation while maintaining authenticity and resolution. These methods are categorized into label-guided and text-guided synthesis.

3.2.1 Label-guided Image/Video Synthesis. Label-guided methods utilize specific labels or annotations to dictate the characteristics of generated portraits. These labels can encompass a wide range of attributes such as age, gender, facial expression, or detailed features like makeup and accessories.

By incorporating these labels into the generative model, the model learns to synthesize portraits that adhere to the specified attributes. Lu et al. [113] used an attribute-guided conditional CycleGAN to generate high-resolution face images from low-resolution inputs, controlled by user-defined attributes. Liu et al. [110] focused on age attributes, presenting a wavelet-based GAN model that captures age-related texture details across multiple scales in frequency space, effectively addressing challenges in face aging with unpaired datasets and mitigating unnatural facial attribute changes. Men et al. [122] demonstrated the Attribute-Decomposed GAN, which embeds human attributes such as pose and clothing into the latent space as independent codes, enabling flexible control through style-based mixing and interpolation. Celeb-HQ [233] is a widely used dataset for face generation, meticulously annotated with 83 facial attributes, benefiting label-guided techniques significantly. Zou et al. [235] introduced a generative framework for creating 3D facial expression sequences using a **Denoising Diffusion Probabilistic Model (DDPM)**, which supports both unconditional and conditional training with various inputs such as expression labels and partial sequences.

Label-guided portrait synthesis offers precise control over generated image attributes, benefiting applications requiring specific visual features and enhancing data efficiency by reducing the need for extensive variant datasets. However, these methods can limit creative freedom and relies heavily on accurately labeled training data, which can introduce biases. To overcome these limitations, researchers have turned to text-guided synthesis, which uses textual descriptions for greater flexibility and control.

3.2.2 Text-guided Image/Video Synthesis. Text-guided methods extend the abilities of label-guided techniques by providing nuanced and detailed control over generated images through natural language descriptions. These methods use text descriptions to specify complex combinations of features and scenarios in a more expressive and flexible manner. Mansimov et al. [120] made the first attempt at image generation from natural language, employing a combination of RNN and VAE, but the results were blurry. Reed et al. [141] pioneered the use of GANs for synthesizing realistic images from text descriptions, combining advanced **recurrent neural networks (RNNs)** for text feature extraction with deep convolutional GANs for image generation. Xu et al. [201] presented an attention-driven GAN for detailed text-to-image synthesis, focusing on relevant words in text descriptions to allow precise control over fine-grained image details. Canfes et al. [19] enabled control over both shape and texture using text or image-based prompts by manipulating the latent space of 3D generative models with the help of the CLIP model and a pre-trained 3D GAN model. ARG have also been employed for text-guided image generation using large-scale data. Ramesh et al. [140] introduced DALL-E, a transformer that models text and image tokens as a unified data stream, simplifying the process by treating text and image information equally and performing competitively with sufficient data and scale. Yu et al. [213] proposed Parti, which treats the task as a sequence-to-sequence problem similar to machine translation, using a Transformer-based image tokenizer to encode images into discrete tokens. Both Parti and DALL-E suffer from high computation costs due to their large number of parameters.

More recently, diffusion models have made major breakthroughs in this task. Huang et al. [78] detailed Collaborative Diffusion, which enables multimodal control over the generative process in diffusion models. This approach leverages existing pre-trained uni-modal diffusion models, facilitating multimodal face generation and editing without the need for re-training. Wu et al. [194] developed an advanced neural rendering-based approach to text-to-3D-portrait generation with 3DPortraitGAN, producing high-quality 3D portraits consistent with textual descriptions. The quality of text-to-video generation has also improved significantly. Ho et al. [67] presented Imagen Video, an advanced text-conditional video generation system utilizing a cascade of video

diffusion models to generate high-definition videos. Wu et al. [192] introduced One-Shot Video Tuning for text-guided video generation, which requires only a single text-video pair for training. Tune-A-Video employs a spatiotemporal attention mechanism and an efficient one-shot tuning strategy to capture continuous motion, producing smooth and coherent videos.

Text-guided generation is effective for creating images that match text descriptions, with advancements like **Adaptive Variational Parameters (AVP)** and **Actor-Score Distillation Sampling (ASDS)** enhancing 3D portrait generation. However, it faces limitations such as reliance on large pre-trained datasets, biases in training data, high computational resource demands, and poor generalization to unseen text descriptions. Building on the foundation of portrait generation, we now turn to talking head driving mechanisms, focusing on generating realistic talking head videos through various driving methods.

4 Driving Mechanisms

Driving mechanisms are pivotal in advancing the field of talking head synthesis, enabling the creation of realistic and expressive talking head videos. By leveraging various input modalities, these mechanisms control the facial movements and expressions of synthesized characters, ensuring natural and coherent results. The field can be broadly divided into two main approaches: video-driven methods and audio-driven methods.

4.1 Video-driven Methods

Video-driven generation aims to create realistic face videos by preserving the identity of a still image while replicating the motion from a driving video. High-resolution video generation requires capturing finer facial details and ensuring temporal consistency, which demands larger datasets and longer training times. Video-driven methods are categorized into two main types: traditional non-learning methods, which use predefined algorithms or manual processes, and advanced learning-based methods, which employ deep learning to generate realistic and expressive facial animations from extensive datasets.

4.1.1 Traditional Parametric Methods. Before the advent of deep learning technologies, accomplishing video-driven tasks such as facial animation from video input was a formidable challenge. Early animation techniques [32, 134], including rule-based methods and conventional image processing, made it feasible to track and animate facial features effectively. These traditional methods largely relied on geometric transformations, template-based modifications, and hybrid transformations.

Geometric Transformations involve using mathematical formulas to manipulate facial features by applying transformations like rotation, scaling, and translation. This approach is straightforward and computationally efficient, allowing for the alteration of facial expressions based on predefined rules or sequences. **Template-based Modifications** involve adjusting predefined facial templates to match the movements observed in a source video. This method typically uses a standard set of facial templates, such as muscle-based models [118, 190] and blendshape models (a predefined facial action template). **Hybrid Transformations** combine traditional transformations with deep learning techniques to achieve video-driven talking head synthesis. Thies et al. [168] introduced a real-time facial reenactment method using monocular video streams, where dense photometric consistency measurements track facial expressions across source and target videos, facilitating reenactment through rapid and efficient deformation transfer with meticulous fitting of the inner mouth region. X2Face [191] employs deep learning techniques but fundamentally relies on geometric transformations to adjust a source face based on a driving face or other input modalities, achieving sophisticated video and image editing.

Table 2. Summary of Advanced Learning-based Methods of Video-driven Branch

Method	Publication		Model		Characteristic		
	Year	Venue	Unsupervised	End-to-end	N-shot	3D model	Driven region
Explicit Modeling Paradigm							
MarioNETte [216]	2019	ICCV	✓		Few-shot	3D	Face
FOMM [153]	2019	NeurIPS	✓	✓	One-shot	2D	Whole head
MeshG [208]	2020	ACM MM	✓	✓	One-shot	3D	Face
Face-vid2vid [183]	2021	CVPR	✓		One-shot	3D	Whole head
SAFA [179]	2021	3DV	✓	✓	One-shot	3D	Face
TPSM [227]	2022	CVPR	✓	✓	One-shot	2D	Whole head
DaGAN [70]	2022	CVPR	✓		One-shot	2D	Whole head
Dual-GFR [72]	2022	CVPR			One-shot	3D	Face
ROME [89]	2022	ECCV	✓	✓	One-shot	3D	Whole head
Free-headGAN [41]	2023	TPAMI		✓	Few-shot	3D	Whole head
Metaportrait [217]	2023	CVPR			One-shot	2D	Whole head
Head3D [127]	2024	WACV	✓		One-shot	3D	Whole head
GONHA [100]	2024	NeurIPS			One-shot	3D	Whole head
PTUS [103]	2024	AAAI			One-shot	3D	Whole head
AnyTalk [186]	2025	AAAI		✓	One-shot	3D	Whole head
HYPortrait [203]	2025	CVPR			One-shot	2D	Whole head
Implicit Modeling Paradigm							
ReenactGAN [193]	2018	ECCV			One-shot	2D	Whole head
Fast-bi [215]	2020	ECCV		✓	One-shot	2D	Whole head
LSR [123]	2021	ICCV		✓	Few-shot	2D	Whole head
FaceGAN [170]	2021	WACV		✓	One-shot	3D	Face
DNRF [54]	2021	CVPR	✓	✓	Few-shot	3D	Whole head
Image-Set [119]	2022	NeurIPS	✓	✓	Few-shot	2D	Whole head
LIA [187]	2022	ICLR	✓		One-shot	2D	Whole head
LNFR-GAN [15]	2022	BMVC	✓		One-shot	3D	Face
TS-Net [128]	2023	WACV	✓		Few-shot	2D	Whole head
ETHVE-S2LS [130]	2023	ICCV		✓	One-shot	3D	Face
Hyperreenact [16]	2023	ICCV		✓	One-shot	3D	Face
HiDeNeRF [99]	2023	CVPR			One-shot	3D	Whole head
Megactor [205]	2024	ArXiv		✓	One-shot	2D	Whole head
LDC [206]	2024	AAAI		✓	One-shot	3D	Whole head

“Year” refers to the publication year. “Venue” indicates the conference or journal where the study was published.

“Unsupervised” refers to methods that do not require labeled data for training. “End-to-end” describes methods where the entire process from input to output is learned within a single framework, without the need for manual feature engineering. “N-shot” indicates that the input requires either one image (one-shot) or multiple images (few-shot). “3D Model” signifies the use of a 3D model in the method, such as 3DMM or FLAME, “3D” indicates the use of a 3D model, while “2D” indicates its absence. “Driven Region” denotes the area that can be manipulated by driving.

Non-learning methods face significant limitations due to their lack of adaptability and scalability, requiring extensive manual setup and fine-tuning for each new face or expression. They struggle to capture the complexity of human facial expressions because of their reliance on rigid templates or simplistic geometric rules. In contrast, learning-based methods automate facial animation and improve accuracy by learning from vast amounts of data, capturing nuances with greater fidelity.

4.1.2 Advanced Learning-based Methods. Advanced learning-based methods have revolutionized video-driven talking head synthesis by using deep learning to create highly realistic and expressive facial animations. These techniques dynamically learn complex mappings from extensive datasets, significantly enhancing animation quality and accuracy. The approaches are broadly categorized into two paradigms: explicit modeling and implicit modeling. We summarize advanced learning-based methods of video-driven branch from multiple perspectives in Table 2.

Explicit modeling paradigm in video-driven talking head synthesis involves the direct manipulation of clearly defined features and parameters, such as keypoints and meshes, making these

methods highly interpretable and often simpler to implement. Keypoints-based warping methods generate motion flow by learning the correspondence between keypoints, thereby warping the features of the source image [152, 154]. Siarohin et al. [153] developed the **First Order Motion Model (FOMM)**, a method that animates objects in images using keypoints derived via self-supervised learning and first-order local affine transformations, generating a dense motion field and occlusion masks to deform the source image at the encoder's feature layer and restore it through the decoder. Zhao et al. [227] improved upon FOMM by predicting multiple sets of keypoints for thin-plate spline transformations and incorporating an affine transformation for background prediction. DA-GAN [70] further enhanced FOMM by utilizing depth information to improve the precision of warps and reduce artifacts. Face vid2vid [183] introduced dynamic 3D keypoints for unsupervised learning from single-source images, enabling one-shot free-view avatar synthesis. Zhang et al. [217] leveraged dense landmarks to generate precise geometry-aware flow fields, ensuring the adaptive fusion of source identity during synthesis to maintain essential features. Doukas et al. [41] used sparse 3D facial landmarks for face modeling, eliminating the need for complex facial priors, and incorporated explicit gaze control and an attention mechanism within the generator to facilitate few-shot learning and accommodate multiple source images. To address challenges like inadvertent transfer of the driver's identity and handling unfamiliar large poses, Ha et al. [216] implemented image attention blocks, target feature alignment, and landmark transformers, with the landmark transformer playing a critical role in preserving identity by decoupling landmarks to isolate expression geometry from identity traits.

In contrast to keypoints-based methods, mesh-based rendering methods depend on 3D head reconstruction models and 3D representation models to provide a more comprehensive understanding of human facial dynamics. These models offer a detailed perspective for reenacting facial movements [23]. Besides the widely used 3DMM, models like DECA [52], Emoca [39] are also employed. Yao et al. [208] proposed a method that uses 3D mesh reconstruction to guide optical flow learning for facial reenactment. Wang et al. [179] utilized 3DMM for facial simulation while applying multiple affine transformations to render foreground elements like hair and beard. Khakhulin et al. [89] introduced ROME, a mesh-based method for creating one-shot human head avatars from a single photo, determining the head mesh and neural textures from the photograph. Li et al. [100] presented an innovative approach for reconstructing and animating 3D head avatars from single portrait images, using a tri-plane method [21] to capture aspects of 3D geometry, appearance details, and facial expressions.

Keypoints-based warping methods maintain subject identity but struggle with occlusion and significant motion, while mesh-based rendering offers higher fidelity but faces challenges like model complexity and high computational demands. These limitations lead to the implicit modeling paradigm, which uses latent embeddings to infer features and relationships, offering greater flexibility and handling complex, non-linear interactions in dynamic facial animations.

Implicit Modeling Paradigm in video-driven talking head synthesis utilizes latent embeddings or spaces to infer features and relationships, providing greater flexibility and handling complex, non-linear interactions. Latent space-based methods represent images as embedding codes. Wu et al. [193] introduced ReenactGAN, the first end-to-end learning-based facial reenactment framework, which translates the source face into a boundary latent space, adjusts it with a transformer, and reconstructs the target frame using a target-specific decoder. Wang et al. [187] proposed the LIA model, achieving animation by learning orthogonal motion directions within the latent space and linearly combining them, avoiding complex processing based on structural representation. Meshry et al. [123] developed a few-shot face synthesis method that decomposes the target subject's representation into spatial and style components, predicting the latent space layout and using it for spatial denormalization to synthesize the target frame. Zakharov et al. [215] presented a facial

rendering technique based on neural networks, hierarchically modeling facial features within an embedding space by processing low-frequency information and high-frequency texture separately. Ni et al. [128] introduced a dual-branch network architecture that captures motion from driving videos and creates high-quality video content. Bounareli et al. [15] addressed the separation of identity and posture by combining a pre-trained GAN with a 3D shape model, manipulating facial orientations and expressions within the GAN's latent space. Oorloff et al. [130] utilized StyleGAN2 image inversion and multi-stage non-linear latent space editing to reproduce high-resolution facial videos. Bounareli et al. [16] further refined this approach by employing a hypernetwork to fine-tune source identity features, addressing visual artifacts and reducing fine-tuning costs.

NeRF methods also play a crucial role in video-driven tasks. Gafni et al. [54] demonstrated a dynamic NeRF for simulating facial appearance and movement, combining scene representation networks with a low-dimensional deformable model for explicit control over posture and expressions, using volumetric rendering techniques. Hong et al. [71] presented HeadNeRF, a parametric head model using NeRFs to render high-fidelity human head images, incorporating 2D neural rendering and improving detail accuracy. Li et al. [99] introduced HiDe-NeRF, which decomposes 3D dynamic scenes into canonical appearance and implicit deformation fields, maintaining facial shape and details through multi-scale volumetric features. Other implicit methods include Tripathy et al. [170], who conveyed facial actions through **Action Unit (AU)** representation to prevent identity information leakage. Mallya et al. [119] employed an implicit warping approach for motion transfer, using a cross-modal attention layer to generate image animation by creating correspondences between source and driving images. Hong et al. [69] used implicit identity representation to better preserve the source image's identity. However, this lack of explicit structure can reduce model interpretability, complicating fine-tuning and debugging.

Video-driven methods have advanced the generation of realistic talking head animations by preserving the identity of a still image while replicating the motion from a driving video. However, they face challenges such as occlusion, maintaining identity integrity, and handling large pose variations. In contrast, audio-driven methods leverage audio signals to synchronize lip movements and facial expressions with spoken content, thereby enhancing the realism and naturalness of talking head animations.

4.2 Audio-driven Methods

Audio-driven talking head synthesis transforms a static facial image into a dynamic video synchronized with audio inputs. This multimodal task faces several challenges due to the inherent differences between audio and visual modalities, such as lip synchronization [112] and the diversity and authenticity of expressions [45]. To address these challenges, existing approaches are classified into two main categories: statistical parametric models and cutting-edge deep generative models. We also systematically summarize audio-driven methods in Table 3.

4.2.1 Statistical Parametric Models. The evolution of statistical parametric models for audio-driven talking head synthesis can be traced through the development and application of **Hidden Markov Models (HMMs)**, **Convolutional Neural Networks (CNNs)**, and **RNNs**.

HMMs-based facial animation methods effectively capture the dynamics in video and audio sequences [30, 31]. Yamamoto et al. [204] pioneered the use of HMMs in audio-driven animation by developing a method to synthesize lip movements from voice input, effectively addressing coarticulation challenges in lip movement prediction. Subsequent methods based on HMMs evolved to improve upon these initial efforts [5, 93]. However, due to the lack of a one-to-one correspondence between phonemes. Xie et al. [197] introduced the **coupled hidden Markov model (CHMM)**, a multi-stream approach that enhances animation realism by modeling the asynchrony and temporal

Table 3. Summary of Audio-driven Methods

Method	Publication		Model		Characteristic		
	Year	Venue	Unsupervised	End-to-end	N-shot	3D model	Driven region
Statistical Parametric Models							
CHMM [197]	2007	PR			N/A	2D	Mouth
VL-DNN [223]	2013	Interspeech			N/A	2D	Mouth
PR-BLSTM [48]	2015	ICASSP			N/A	2D	Face
Synthesizing Obama [161]	2017	TOG			N/A	2D	Mouth
VOCA [36]	2019	CVPR		✓	N/A	3D	Whole head
You said that [79]	2019	IJCV	✓		Few-shot	2D	Face
DAVD-Net [224]	2020	CVPR		✓	Few-shot	2D	Face
MakeItTalk [231]	2020	TOG		✓	One-shot	2D	Whole head
NVP [166]	2020	ECCV		✓	N/A	3D	Whole head
Everybody's token [155]	2022	TIFS		✓	N/A	3D	Whole head
DiNET [225]	2023	AAAI			One-shot	2D	Face
Cutting-edge Deep Generative Models							
DAVS [229]	2019	AAAI		✓	Few-shot	2D	Face
ATVG [26]	2019	CVPR		✓	One-shot	2D	Face
Wav2Lip [138]	2020	ACM MM			Few-shot	2D	Face
RS-DFA [172]	2020	IJCV		✓	One-shot	2D	Face
DAR [125]	2020	WACV		✓	N/A	2D	Face
FACIAL [220]	2021	ICCV		✓	Few-shot	2D	Whole head
AG-Codec [143]	2021	WACV			Few-shot	2D	Face
AD-NeRF [61]	2021	ICCV			Few-shot	3D	Whole head
FaceFormer [50]	2022	CVPR	✓	✓	N/A	3D	Face
DFA-NeRF [209]	2022	ArXiv	✓		Few-shot	3D	Whole head
DFRF [150]	2022	ECCV			Few-shot	3D	Whole head
PAV-DP [214]	2023	ICCV			One-shot	2D	Whole head
EMMN [164]	2023	ICCV			Few-shot	2D	Face
FONT [104]	2023	ICME	✓		One-shot	2D	Whole head
SadTalker [222]	2023	CVPR	✓	✓	One-shot	2D	Whole head
LipFormer [177]	2023	CVPR		✓	Few-shot	2D	Face
ER-NeRF [97]	2023	CVPR			N/A	3D	Whole head
GeneFace [210]	2023	ICLR			Few-shot	3D	Whole head
DiffTalk [151]	2023	CVPR			Few-shot	2D	Whole head
Dreamtalk [116]	2023	ArXiv	✓		One-shot	3D	Whole head
TGDM [200]	2022	ArXiv			One-shot	3D	Whole head
ACDM [12]	2024	IVC		✓	One-shot	2D	Whole head
MusicFace [108]	2024	CVM			N/A	3D	Whole head
SyncTalk [136]	2024	CVPR	✓	✓	Few-shot	3D	Whole head
Diffused Heads [156]	2024	WACV			One-shot	2D	Whole head
Two-Stage [11]	2024	PROPOR			N/A	2D	Whole head
V-Express [174]	2024	ArXiv			Few-shot	2D	Whole head
Talkinggaussian [96]	2024	ECCV		✓	Few-shot	3D	Whole head
PointTalk [198]	2025	AAAI		✓	Few-shot	3D	Whole head
SyncTalk++ [135]	2025	ArXiv		✓	Few-shot	3D	Whole head
HiCoDe [195]	2025	AAAI		✓	Few-shot	3D	Whole head

“Year” refers to the publication year. “Venue” indicates the conference or journal where the study was published.

“Unsupervised” refers to methods that do not require labeled data for training. “End-to-end” describes methods where the entire process from input to output is learned within a single framework, without the need for manual feature engineering. “N-shot” indicates that the input requires either one image (one-shot) or multiple images (few-shot). “3D Model” signifies the use of a 3D model in the method, such as 3DMM or FLAME, “3D” indicates the use of a 3D model, while “2D” indicates its absence. “Driven Region” denotes the area that can be manipulated by driving.

dependencies of audiovisual speech, converting acoustic speech into high-quality facial animation parameters using an **Expectation Maximization (EM)** algorithm.

CNN-based facial animation methods utilize deep learning architectures to enhance the quality and realism of facial animations. Chung et al. [79] pioneered the use of CNNs to synthesize animated talking face videos from a single image and its corresponding audio track, presenting an encoder-decoder framework that merges facial and audio data embeddings. With the development of deep learning, Karras et al. [84] proposed a DCNN method that incorporates deformable convolutions for

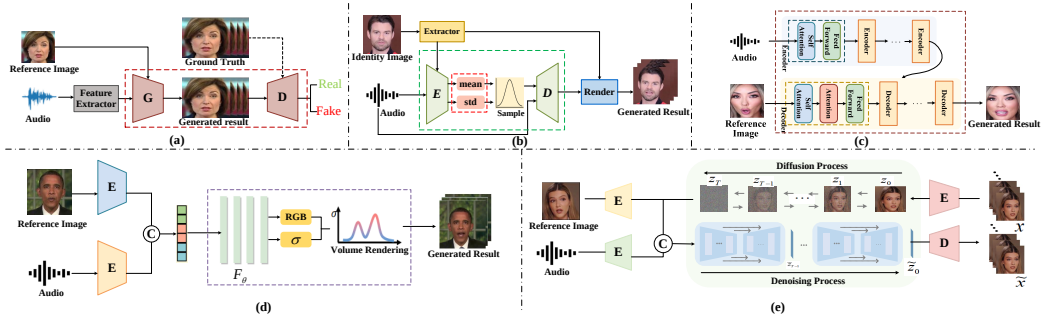


Fig. 4. Typical structures for cutting-edge deep generation models of audio-driven branch. (a) GAN-based audiovisual synthesis methods. (b) VAE-based audiovisual mapping methods. (c) Transformer-based sequence translation. (d) Implicit field animation modeling. (e) Diffusion-based expression synthesis methods.

reconstructing talking head videos at extremely low bitrates, embedding encoder information from video compression standards and using a constrained projection module to repair compression artifacts, improving reconstruction quality. Cudeiro et al. [36] introduced VOCA, a neural network that generates realistic 3D facial animations from speech signals across languages, incorporating diverse speaking styles and identity-specific characteristics without retraining. Zhang et al. [225] introduced DiNet, a method for few-shot learning in high-resolution facial dubbing, using spatial deformation of feature maps to retain texture details and synchronize lip movements with audio. **RNN-based** facial animation methods further refine the ability to handle sequential data, with **Long Short-Term Memory (LSTM)** architectures being the most frequently employed variant [44, 49, 166]. Fan et al. [48] used deep **Bidirectional Long Short-Term Memory (BLSTM)** networks for audio-visual modeling, converting data into parallel time series and training the BLSTM to minimize errors between context labels and visual features for more detailed animation. Suwajanakorn et al. [161] trained a neural network on audio clips of President Obama to map audio features to mouth shapes, synthesizing high-quality lip textures into target videos using 3D pose matching. Song et al. [155] proposed a method that decomposes video frames into expression, geometry, and pose parameters, using a recurrent network to transform audio into synchronized mouth movements while preserving geometric and postural context. Zhou et al. [231] introduced a method for generating expressive talking head videos from a single facial image and audio input by separating lip movements and other dynamics, and predicting facial landmarks to reflect the speaker's intentions. The above methods have significantly advanced audio-driven facial animation by enhancing realism and handling sequential data. However, they face limitations in maintaining lip synchronization, achieving diverse and authentic expressions, and personalizing the speaker's style. Cutting-edge deep generative models leverage sophisticated neural architectures and extensive datasets to produce superior results in audio-driven talking head synthesis, addressing these challenges effectively.

4.2.2 Cutting-edge Deep Generative Models. The advent of cutting-edge deep generative models has revolutionized audio-driven talking head synthesis by leveraging advanced neural architectures and extensive datasets to enhance lip synchronization, achieve diverse expressions, and personalize the speaker's style. Key approaches include GAN-based, VAE-based, transformer-based, implicit field-based, explicit parametric-based, and diffusion-based, each introducing unique innovations that advance realistic and expressive talking head animations. Figure 4 compares the main structure of each key approach.

GAN-based audiovisual synthesis methods harness the capabilities of GANs to seamlessly integrate auditory cues with visual data [172, 220], enabling precise and realistic facial animations. Chen et al. [26] developed a cascaded GAN that converts audio into facial landmarks before generating video frames, using attention and sequence-aware discriminator to remove pixel jittering and avoid spurious correlations. Zhou et al. [229] developed a method to generate adaptable speaking faces for arbitrary subjects by decoupling subject-specific and speech-related information using joint adversarial training. Prajwal et al. [138] introduced Wav2Lip, which enhances lip synchronization across various identities in dynamic contexts and establishes rigorous benchmarks for future research, including emotional and music-driven synthesis. Tan et al. [164] propose EMMN, synthesizing speaking facial expressions by combining emotional embeddings and lip movements, ensuring consistent facial expression synthesis. Liu et al. [108] address music-driven singing facial synthesis with a method for natural facial movements and a SingingFace dataset for research. These methods offer substantial improvements in realism and adaptability for talking head synthesis. While they excel in producing high-quality animations and maintaining lip synchronization, challenges such as avoiding spurious correlations and ensuring expression consistency persist.

VAE-based audiovisual mapping methods excel in capturing the complex inter-dependencies between audio and visual modalities, generating lifelike and expressive facial animations. Mittal et al. [125] introduce an audio representation learning framework that segregates speech content, emotional tone, and background noise, resulting in more precise lip movements even amidst disruptive elements. Liu et al. [104] propose a flow-guided one-shot model that generates talking head videos with natural head movements by integrating head pose prediction based on a CVAE probabilistic model, unsupervised keypoint extraction, and flow-guided occlusion-aware video synthesis techniques, effectively bridging the audio-visual modality gap. Richard et al. [143] demonstrate a real-time 3D facial model that accurately predicts facial coefficients representing geometric and texture features using latent cues from lossy input signals like audio and eye-tracking, based on a multimodal VAE. Zhang et al. [222] introduce SadTalker, which synthesizes 3D motion coefficients from audio inputs and integrates them with advanced 3D perception rendering technology to accurately map audio-motion correlations, capturing detailed facial expressions and head movements. However, their reliance on high-quality training data can lead to overfitting or poor generalization if the data is insufficient or biased. Additionally, their stochastic nature can result in inconsistent animations over longer sequences.

Transformer-based sequence translation methods leverage the capabilities of the Transformer model, initially proposed and widely adopted in the field of **Natural Language Processing (NLP)** to address long-distance dependency issues and improve sequence-to-sequence tasks [1, 8, 63]. Fan et al. [50] leveraged a transformer-based autoregressive model with a tailored biased attention mechanism to capture long-term audio context, aligning audio-motion modalities, and accurately forecasting animated 3D facial mesh sequences for enhanced lip movement precision. Huang et al. [77] used a transformer-based encoder to map audio signals to 3DMM parameters, guiding the generation of high-quality avatars by predicting long-term audio context. Wang et al. [177] presented a method for synthesizing high-fidelity talking heads by using a pre-trained facial image codebook within the Transformer framework, predicting sequences of lip codes from audio features to accurately mimic speech-related lip variations. Bernardo et al. [11] introduced a dual-stage framework combining transformers to extract facial landmarks from audio and GANs to generate realistic video frames, allowing for dynamic shapes and multiple appearances by adjusting weights. However, these methods require substantial computational resources and extensive datasets to achieve optimal performance.

Implicit field-based animation modeling methods utilize advanced 3D reconstruction models to generate highly realistic talking head animations. A prominent approach in this category is

NeRF, initially proposed by Mildenhall et al. [124], which is particularly relevant to this task. Guo et al. [61] used neural scene representation networks to generate dynamic neural radiance fields from audio signals, synthesizing high-fidelity talking head videos with flexible adjustments to audio, viewing angles, and backgrounds, and extending to upper body generation. Yao et al. [209] employed a neural radiance field to decouple lip motion from personalized features, predicting lip movements from audio and sampling head movements and blinking through a VAE. Shen et al. [150] introduced the **Dynamic Facial Radiance Field (DFRF)**, which uses 2D appearance images and a differentiable facial deformation module to quickly adapt to new identities and generate high-quality talking heads with minimal training data. Li et al. [97] adopted conditional neural radiance fields with NeRF-based tri-plane hash representations and spatial regions to generate talking portraits, achieving rapid convergence, real-time rendering, and high efficiency. Ye et al. [210] proposed a method for generating realistic video portraits from arbitrary audio, using a VAE and a conditioned NeRF renderer to address head-body disconnection, enhancing video fidelity and generalizability. These methods offer flexibility in adjusting audio signals, viewing angles, and backgrounds, thereby enhancing the realism and adaptability of the generated animations. However, they often require extensive computational resources and sophisticated models to achieve optimal results.

Explicit parametric-based modeling methods leverage the strengths of explicit representations to address the limitations of implicit radiance field models. The latest technique, 3D Gaussian, demonstrates superior performance in 3D reconstruction compared to NeRF [232]. First, we introduce some research on 3D human head avatar modeling. Xu et al. [202] utilized controllable 3D Gaussian models for high-fidelity head avatar modeling. By incorporating high-frequency dynamic details through a fully learned MLP-based deformation field, it effectively simulates a wide range of extreme expressions. Zhao et al. [228] demonstrated that PSAvatar uses a **Point-based Morphable Shape Model (PMSM)** with 3D Gaussian modeling, excels in real-time animation by enabling flexible and detailed 3D geometry modeling. Cho et al. [29] proposed GaussianTalker, a real-time pose-controllable talking head model that integrates 3D Gaussian attributes with audio features into a shared implicit feature space, leveraging **3D Gaussian Splatting (3DGS)** for rapid rendering, resulting in enhanced facial fidelity, lip-sync accuracy, and superior rendering speed compared to existing models. Li et al. [96] presented a unified audio-driven talking head synthesis framework built upon 3DGS. The method leverages a dual-branch architecture: a static branch encodes the identity and head geometry via a canonical Gaussian field, while a dynamic branch modulates lip motion based on audio features. Xie et al. [198] constructed a static 3D Gaussian field for the head and drives deformation via an audio-conditioned dynamic lip point cloud. A dynamic difference encoder captures subtle lip motions, while an audio-point enhancement module aligns and fuses cross-modal features. This Gaussian-centric design enables high-fidelity rendering and superior audio-lip synchronization, achieving real-time performance with strong generalization. Peng et al. [135] proposed a 4D Gaussian Splatting framework tailored for precise lip-audio synchronization in talking head generation. By explicitly modeling temporal dynamics within the Gaussian field, the method enables smooth transitions and accurate facial motion. This spatio-temporal Gaussian representation offers both high visual fidelity and precise lip-sync alignment. However, the parametric nature of 3D Gaussians imposes limitations on the complexity of shapes that can be modeled without a substantial increase in the number of Gaussians, thereby escalating computational costs and potentially affecting efficiency.

Diffusion-based expression synthesis methods leverage diffusion models to incrementally add and remove noise from data, transforming it to and from a Gaussian distribution to generate high-quality images [58, 146]. Yu et al. [214] achieved realistic and natural lip-sync by sampling facial movements that are unrelated to lip movements, leveraging the probabilistic characteristics

Table 4. Summary of Editing Techniques in Talking Head Synthesis

Method	Year	Venue	Driven Content	Whole-frame Manipulation	Controllable Attribute			
					Lip Sync	Head Pose	Expression	Eye Blink
2D Image/Video Editing								
ICface [169]	2020	WACV	Image	✓		✓	✓	✓
PC-AVS [230]	2021	CVPR	Audio and video		✓	✓	✓	
GC-AVT [102]	2022	CVPR	Audio and image		✓	✓	✓	
DPE [132]	2023	CVPR	Video	✓		✓	✓	✓
PDRL [175]	2023	CVPR	Audio and image		✓	✓	✓	✓
OPT [105]	2023	ICASSP	Audio and image	✓	✓	✓	✓	
EDTalk [163]	2024	ECCV	Audio	✓	✓		✓	
EchoMimic [27]	2025	AAAI	Audio	✓	✓	✓	✓	✓
PC-Talk [173]	2025	ArXiv	Audio and text	✓	✓	✓	✓	✓
3D Model Editing								
TBEHead [53]	2019	TOG	Text	✓	✓	✓		✓
3DTF-PPD [218]	2021	TCVG	Audio	✓	✓	✓		
HeadGAN [42]	2021	ICCV	Video	✓		✓	✓	
PIRenderer [142]	2021	ICCV	Video or audio	✓	✓	✓	✓	
NED [133]	2022	CVPR	Video	✓		✓	✓	✓
OTAvatar [117]	2023	CVPR	Video	✓		✓	✓	
PECHHead [55]	2023	CVPR	Video	✓		✓	✓	
Styletalk [115]	2023	AAAI	Audio and video	✓	✓	✓	✓	
TalkCLIP [114]	2023	ArXiv	Audio and text	✓	✓	✓	✓	
LES-Talker [51]	2024	ArXiv	Audio and image	✓	✓	✓	✓	✓
Control-Talker [101]	2024	ACM MM	Audio and image	✓	✓	✓	✓	✓
EmotiveTalk [176]	2025	CVPR	Audio	✓	✓		✓	

“Year” refers to the publication year. “Venue” indicates the conference or journal where the study was published. “Driven Content” denotes the source of driving input, including video or audio. “Whole Frame Manipulation” describes the method’s focus on generating the entire image, encompassing both the person and the background. “Controllable Attribute” outlines the specific features that can be manipulated, such as Lip Sync, Head Pose, Expression, and Eye Blink.

of audio-to-visual diffusion priors to produce varied facial movement sequences for the same audio input. Shen et al. [151] used latent diffusion models to generate high-resolution, audio-synchronized talking head videos that adapt to new identities without fine-tuning. Ma et al. [116] presented a diffusion model framework that predicts target expressions from audio and generates realistic, style-varied talking heads with precise lip movements. Xu et al. [200] introduced the **Texture-Geometry-aware Diffusion Model (TGDM)** that utilizes the CLIP model for nuanced emotional control, employing a multi-conditional denoising task to produce highly realistic talking faces. Stypulkowski et al. [156] introduced a solution for talking-face generation using autoregressive diffusion models, generating realistic talking head videos from a single identity image and audio sequence. Bigioi et al. [12] demonstrated an end-to-end audio-driven video editing method using denoising diffusion models to generate synchronized facial movements directly from audio Mel spectrogram features, eliminating the need for intermediate representations like facial landmarks or 3D models. These advancements underscore the efficacy of diffusion-based approaches in producing lifelike, expressive talking heads, setting a new benchmark in audiovisual synthesis.

5 Editing Techniques

Editing techniques in the context of talking head synthesis build upon the foundational phases of portrait generation and talking head animation. These techniques are essential for refining and customizing the final outputs to meet specific needs and enhance realism. Editing techniques can be broadly classified into two categories: 2D image/video editing and 3D model editing. We summarize common methods of editing in Table 4.

5.1 2D Image/Video Editing

2D image/video editing for talking heads involves advanced techniques that enhance and modify facial animations by manipulating specific visual attributes while maintaining overall video

coherence, divided into two primary approaches: disentanglement and decoupling, and control and modulation.

5.1.1 Disentanglement and Decoupling. These techniques focus on separating and independently controlling various aspects of facial animations in talking heads. By isolating different attributes such as expression, identity, and pose, these methods enable precise and flexible editing without affecting other visual components. Liang et al. [102] present a method for granular control of a speaking head by decomposing the reference driving image into a cropped mouth, a masked head, and the upper face, achieving a blend of lip synchronization, pose control, and emotional expression. Pang et al. [132] presented a self-supervised decoupling framework that operates independently of 3DMM and paired data. This framework employs a motion editing module to decouple and independently control pose and expression in the latent space, making it suitable for video editing. Wang et al. [175] introduced a method for fine-grained control over lip movement, blink, head pose, and emotional expression by progressively decoupling motion features within unstructured video data, effectively leveraging and separating these motion factors. Tan et al. [163] introduced EDTalk, which accommodates diverse input modalities and flexible control by dividing facial dynamics into three orthogonal latent spaces representing the mouth, pose, and expression, ensuring their independence and allowing for more refined disentanglement and application. Xiong et al. [199] proposed a novel decoupling approach that separates the driving of lip movements from the infusion of image textures, achieving independent control over these two aspects. Audio signals are responsible for driving the movements of the lip contours, while the textural details of the image are handled by a separate network module. In this module, by manipulating style codes of different semantic areas, flexible local facial editing such as hair changes, lip adjustments, or eyebrow styling is possible.

The disentanglement and decoupling techniques have significantly advanced the precision and flexibility of editing talking head animations by isolating and independently controlling various facial attributes such as expression, identity, and pose. These methods offer the advantage of targeted manipulation without compromising other visual components, resulting in highly detailed and customizable animations. However, the complexity of decoupling multiple attributes can introduce challenges in maintaining naturalness and coherence, particularly when dealing with diverse and dynamic input data.

5.1.2 Control and Modulation. Technologies in 2D image and video editing focus on fine-tuning and precisely manipulating specific aspects of facial animations. These methods aim to achieve nuanced control over various facial features, such as lip movements, eye blinks, and emotional expressions, enabling the creation of highly realistic and expressive talking heads. Tripathy et al. [169] proposed a facial animator that controls pose and expressions using human-interpretable signals like head pose angles and AU values, supporting mixed control sources and selective post-editing. Liu et al. [105] designed a network for generating pose-controllable talking heads, using an audio feature disentanglement module to enhance facial structure and identity preservation while enabling free pose control. Zhou et al. [230] presented a framework for pose-controllable talking faces, modularizing audio-visual representations with an implicit low-dimension pose code, effectively operating on non-aligned raw face images. Sun et al. [159] advanced fine-grained controllable facial expression editing by using input AUs to control muscle group intensities and integrating AU features with the expression latent code.

Control and modulation techniques in 2D image and video editing have shown significant advancements in achieving detailed and precise facial animations. These methods offer flexibility in manipulating specific facial features and maintaining the realism of talking heads. However, they often require complex models and extensive data to ensure accuracy and naturalness.

5.2 3D Model Editing

3D model editing techniques represent the forefront of advancements in talking head synthesis, offering unparalleled precision and realism by directly manipulating three-dimensional facial representations. Unlike 2D methods, which often face limitations in capturing complex facial geometries and depth, 3D model editing provides a robust framework for achieving lifelike animations. The field is categorized into two primary approaches: direct parametric control and Multimodal integration.

5.2.1 Direct Parametric Control. These techniques focus on manipulating predefined parameters to achieve precise adjustments in 3D facial models. By leveraging parametric models, they offer fine-grained control over specific facial attributes such as expressions, head poses, and skin textures. Ren et al. [142] introduced a model utilizing 3DMM parameters for controlling facial movements from a single image, adaptable for audio-driven facial reenactment by extracting actions from audio input. Doukas et al. [42] designed a HeadGAN, which edits expressions and poses by adapting 3D facial representations from any driving video to the facial geometry of a reference image, effectively decoupling identity and expressions. Zhang et al. [218] presented a method that uses PoseGAN to generate head poses from audio input and PGFace to create a natural facial model based on these poses. Gao et al. [55] proposed a high-fidelity talking head synthesis method with freely controllable head poses and expressions, combining self-supervised and 3D model-based landmarks with a motion-aware multi-scale feature alignment module to effectively convey motion information. Ma et al. [117] presented OTAvatar, which uses a controllable tri-plane rendering approach to quickly create personalized talking head avatars from a single reference portrait. It converts the portrait into an identity code and merges it with a motion code, effectively separating identity from motion for efficient CNN-based generation and volumetric rendering. Liu et al. [106] introduced a DECA-based facial prior to improve keypoint detection accuracy and incorporated expression encoding, which permits the adjustment and editing of specific expressions during generation, enhancing the model's control over complex facial dynamics. Sun et al. [157] use 3DMM to extract identity and pose parameters from input images, encoding input audio or text into a unified semantic latent space to generate expression parameters. These parameters are then used to guide the deformation and editing of images through a motion field. Li et al. [101] proposed a two-stage method for rapidly customizing talking heads with high-quality texture details. Initially, facial priors are acquired using a pre-trained diffusion model, and then texture details are enhanced through the high-frequency features of images. It only takes 10 minutes to achieve personalized customization with a small number of images, combining with the FLAME-3DMM model to achieve precise control of expressions, poses, and lighting based on audio and video inputs.

Direct parametric control techniques can provide precise adjustments in 3D facial models. However, the reliance on predefined parameters can limit flexibility and the ability to capture more complex, dynamic interactions in facial expressions. This approach often requires extensive manual tuning and can struggle with generalizing to unseen data.

5.2.2 Multimodal Integration. This type involves extracting control information from multimodal inputs and using it to guide the parameterization or adjustment of 3DMM. By leveraging diverse data sources, such as audio, text, and visual cues, these methods create dynamic and contextually appropriate facial animations. Fried et al. [53] developed a method for editing talking head videos by modifying the transcription text to alter video content while maintaining audio-visual fluency. It automatically annotates videos with key parameters, enabling tasks such as adding, deleting, and modifying words, language conversion, and sentence creation. Ma et al. [114] presented a one-shot method for generating talking heads controlled by natural language descriptions. By leveraging a text-video dataset and a CLIP-based style encoder, it aligns textual descriptions with speaking styles, generating realistic talking heads without the need for reference videos. Papantoniou et al. [133]

Table 5. Common Deepfake Detection Methods

Methods	Dataset	Key Techniques	Content Type	Accuracy
MesoNet [2]	DFDC	CNN	Videos	>0.900
Guera et al. [59]	Custom Dataset	RNN	Videos	0.971
Matern et al. [121]	FF	KNN, MLP	Videos and images	0.866
Nguyen et al. [126]	FF	CNN	Videos	0.993
Sabir et al. [145]	FF++	RNN	Videos	0.995

proposed a method for altering an actor's emotional state in a video using only an emotional label to control facial expressions while maintaining lip movements, utilizing 3D facial representation and a deep domain transformation framework. Ma et al. [115] designed a one-shot style-controllable talking face generation framework that extracts speaking style from a reference video and applies it to a single portrait with audio, generating talking head videos with diversified styles. Feng et al. [51] utilized the **Linear Emotion Space (LES)** to achieve fine-grained emotional editing. This method initially defines LES through AUs, then uses attention mechanisms to explore the relationship between LES and 3DMM parameters, enabling controlled deformations of 3D models guided by LES representations. By transforming vectors within LES, corresponding facial editing can be achieved.

Multimodal Integration techniques offer significant advantages in generating contextually appropriate and dynamic facial animations. However, these methods will be limited by the quality and variety of the input data and the complexity of effectively integrating multimodal information, which can impact the consistency and realism of the generated animations.

We have discussed talking head synthesis from static to dynamic in Section 3–5, existing methods enable the creation of natural and high-quality facial images and videos, undoubtedly enhancing its potential applications in entertainment, virtual reality, and other creative domains. However, they also intensify concerns about misuse, particularly in forms of deliberate deception or harmful content. For instance, malicious individuals could exploit these synthetic methods to forge videos and tamper with speeches, thereby creating social unrest, violating personal privacy, or spreading misinformation. These issues seriously impact individuals' social cognition and mental health, heightening concerns over misinformation and undermining trust in authentic media. Therefore, as we enhance the quality and diversity of synthetic outputs, detecting and verifying the generated content becomes equally imperative. Detection methods operate by identifying subtle discrepancies or artifacts in generated content that diverge from real human behavior, including pixel-level details, temporal synchronization, or physiological signals. Traditional methods typically rely on detecting visual or auditory anomalies, while more advanced deep learning-based algorithms utilize large datasets to learn fine-grained patterns of authenticity. The importance of these methods lies in their ability to protect the digital ecosystem which provide reliable tools to detect deceptive videos. Efficient detection methods help protect individual reputations, prevent the spread of misinformation, and maintain societal trust in online media. Table 5 presents a curated list of top deepfake detection methods, by comparing and analyzing these methods, readers can understand the developments in detection research, more detailed information can be found in these latest surveys [82, 88].

6 System-Level Integration: Issues and Research Frontiers

Despite impressive progress within the three canonical stages of talking head synthesis: identity modeling, motion representation, and rendering, most advances remain stage-specific. When these components are fused into an end-to-end pipeline, subtle inter-dependencies surface and trigger failure modes that single-stage metrics cannot capture. We distil the integration problem into five orthogonal yet interacting technical axes, and highlight research frontiers that must be crossed before talking head generation becomes a dependable, real-time technology.

Identity Consistency Across Stages. Large-amplitude head motions, spontaneous affect, and interactive edits exacerbate the accumulation of small representation errors introduced at stage boundaries. Identity latents produced by a portrait generator may no longer align with deformation fields estimated by a motion driver, yielding gradual *identity drift*. Promising remedies include globally shared identity codes, cycle-consistent regularisers, and projection layers that map heterogeneous embeddings into a common manifold while preserving visual fidelity over hundreds of frames.

Unified Representation and Interface Compatibility. A contemporary pipeline operates on at least three distinct vocabularies—dense portrait latents, sparse 2D/3D key-points or meshes, and high-level parametric controls. The lack of a “lingua franca” forces researchers to build brittle, ad-hoc translators, limiting reproducibility, and system reuse. A formally specified, differentiable intermediate representation—e.g., a hybrid graph–tensor structure—combined with bidirectional transformers that offer quantifiable distortion budgets is urgently needed to enable plug-and-play experimentation.

Real-Time Synchronisation and Computational Coordination. Interactive use-cases impose sub-100-ms round-trip latency, yet the three stages differ by orders of magnitude in FLOPs and memory. Blindly chaining them thus yields either under-utilised hardware or unpredictable frame time spikes. Research opportunities lie in latency-aware graph partitioning, asynchronous pipelining, and on-the-fly model scaling that can redistribute compute budget per frame. Complementary directions include learned surrogate modules that cache slow components, and uncertainty-driven adaptive sampling schemes that invoke expensive computation only for perceptually sensitive frames.

Joint Optimisation and End-to-End Training. Stage-wise pre-training maximises local metrics but often degrades global coherence. A principled alternative is to expose a single differentiable graph from audio/text to pixels and to optimise a Pareto-balanced objective spanning identity fidelity, expression realism, temporal smoothness, and efficiency. Open problems include designing gradient-stable cross-stage interfaces, scale-invariant multi-objective schedulers, and curricula that resolve conflicts between early-stage identity preservation and late-stage expressiveness.

Audio-Visual Alignment and Micro-Expression Fidelity. Minute discrepancies between lip motion and spoken audio are readily perceived by viewers, and even subtle exaggerations or omissions of micro-expressions can break immersion. Because audio cues enter the pipeline early whereas final rendering decisions are taken downstream, slight timing mismatches or buffering jitters can desynchronise the two modalities. Promising directions include cross-modal attention networks that continuously propagate phoneme timing to the renderer, emotion-conditioned decoders that fuse prosody with facial AUs, and self-supervised alignment losses derived from the natural coherence of audiovisual signals.

Position of Talking-Head Integration Among Multi-Stage Tasks. Although full-body animation and generic object synthesis follow an *encode–transform–render* template, none impose the simultaneous constraints of identity permanence, phoneme-level articulation, and sub-second latency. These stacked requirements turn talking-head synthesis into a stress-test for multi-stage generative design and reveal a clear gap in existing surveys, which usually treat the three stages in isolation.

In summary, meaningful advances in talking-head synthesis will hinge on the concerted resolution of five system-level bottlenecks: (i) cross-stage identity regularisation that remains effective over minute-long sequences; (ii) a unified, invertible intermediate representation—supported by high-fidelity transformers—that seamlessly bridges identity, motion and rendering modules; (iii) latency-aware orchestration augmented with lightweight surrogate models to guarantee real-time performance; (iv) gradient-stable, end-to-end objectives that strike a Pareto balance among identity

Table 6. Detailed Information of Datasets in Talking Head Synthesis

Dataset	Year	Hour	Speaker	Sentence	Head Movement	Image Size	Environment	URL	Modality	View
GRID [35]	2006	27.5	33	33k	×	360×288, 720×576	Lab	Link	Video, audio	Frontal-view
CREMA-D [20]	2014	11.1	91	12	✓	960×720	Lab	Link	Video, audio	N/A
TCD-TIMIT [64]	2015	11.1	62	6.9k	×	1,920×1,080	Lab	Link	Video, audio	Multi-view
LRW [34]	2016	173	1k+	539k	×	256×256	Wild	Link	Video, text	Frontal-view
MSP-IMPROV [17]	2016	18	12	652	✓	1,440×1,080	Lab	Link	Audio, video, image	Frontal-view
Voxceleb [7]	2017	352	1.2k	153.5k	✓	224×224	Wild	Link	Video	N/A
ObamaSet [161]	2017	14	1	-	✓	N/A	Wild	Link	Video	N/A
MODALITY [37]	2017	31	35	5.8k	×	1,920×1,080	Lab	Link	Audio, video	N/A
Voxceleb2 [33]	2018	2.4k	6.1k	1.1m	✓	224×224	Wild	Link	Video	N/A
LRS2-BBC [3]	2018	224.5	500+	140k+	✓	224×224	Wild	Link	Video, audio, text	Multi-view
LRS3-TED [4]	2018	438	5k+	152k+	✓	224×224	Wild	Link	Video, audio, text	Multi-view
Lombard [6]	2018	3.6	54	5.4k	✓	720×480, 864×480	Lab	Link	Video, audio	Multi-view
MELD [137]	2018	13.7	407	13.7k	✓	N/A	Wild	Link	Video, audio, text	N/A
RAVDESS [111]	2018	7	24	2	✓	1,920×1,080, 1,280×720	Lab	Link	Video, audio	Frontal-view
LRW-1000 [207]	2019	57	2k+	718k	✓	1,024×576, 1,920×1,080	Wild	Link	Video, audio, text	Frontal-view
Faceforensics++ [144]	2019	5.7	1k	1k+	✓	512×512	Wild	Link	Video, image	Frontal-view
FFHQ [86]	2019	-	-	-	×	1,024×1,024	Wild	Link	Image	N/A
MEAD [178]	2020	39	60	20	✓	1,920×1,080	Lab	Link	Video, audio	Multi-view
HDTF [226]	2021	15.8	362	10k	✓	1,280×720, 1,920×1,080	Wild	Link	Video	N/A
TalkingHead-1KH [184]	2021	1000	-	-	✓	N/A	Wild	Link	Video	N/A
CelebV-HQ [233]	2022	65	15653	-	✓	512×512-more	Wild	Link	Video	N/A
Multiface [196]	2022	-	13	-	✓	2,048×1,334, 1,024×1,024	Lab	Link	Image, audio	Multi-view
FaceScape [234]	2023	-	847	-	✓	1,024×1,024	Wild	Link	Image	Multi-view
Renderme-360 [131]	2023	-	500	42	✓	2,448×2,048	Lab	Link	Video, audio	Multi-view
FEED [43]	2024	-	17	-	✓	3,840×2,160	Lab	Link	Video	Multi-view
Emotalk3D [65]	2024	10	30	20	✓	512×512	Lab	Link	Video, audio	Multi-view

fidelity, expressive richness, temporal coherence, and computational efficiency; and (v) robust audio-emotion coupling mechanisms capable of preserving frame-accurate lip-sync while conveying nuanced micro-expressions. Solving these intertwined problems will elevate today's modular prototypes into dependable, deployable talking-head systems.

7 Benchmarking: Datasets, Evaluation Metrics, and Experimental Analysis

7.1 Datasets

With the rapid advancements in deep learning, the field of talking head synthesis has seen significant progress, largely due to the availability of open-source datasets that provide essential data support. We have compiled and summarized detailed information on commonly used and recently released datasets (including three announced in 2022) in Table 6, along with their access links.

For video-driven tasks, the most frequently adopted datasets include Voxceleb [7], Voxceleb2 [33], and TalkingHead-1KH [183]. For audio-driven tasks, the key datasets are CREMA-D [20], LRW [34], and MEAD [178]. As application scenarios expand, the demands for higher image quality, particularly in terms of detail representation and visual effects, have grown. Consequently, several large-scale, high-resolution datasets have emerged in recent years, such as FFHQ [86], HDTF [226], and CelebV-HQ [233]. More importantly, we have gathered comprehensive information for each dataset, including image size, modality, and the view of the subject, which to our knowledge has not been extensively covered in previous reviews of talking head synthesis.

7.2 Evaluation Metrics

A comprehensive evaluation of talking head synthesis models is crucial for understanding their effectiveness. Chen et al. [22] suggested evaluating these models based on four key criteria: identity preservation, visual quality, lip synchronization, and natural motion. While these dimensions are critical, models rarely excel in all areas simultaneously, often focusing on advancements in a single aspect. In the domain of talking head synthesis, both qualitative and quantitative assessments are widely used. While qualitative assessments rely on direct observation and can be subjective, quantitative metrics provide a more objective evaluation. This section introduces several widely recognized quantitative metrics that are crucial for a thorough evaluation of talking head synthesis models.

Structural Similarity (SSIM) [188] is a metric that measures the similarity between two images, aligning with the human visual system's perception of image quality. It evaluates the similarity between the final results and the ground truth based on luminance, contrast and structure. SSIM values range from 0 to 1, where a value closer to 1 indicates lesser distortion and higher image quality. The calculation formula is:

$$\text{SSIM}(X_g, X_r) = \left(\frac{2\mu_{X_g}\mu_{X_r} + C_1}{\mu_{X_g}^2 + \mu_{X_r}^2 + C_1} \right) \cdot \left(\frac{2\sigma_{X_g X_r} + C_2}{\sigma_{X_g}^2 + \sigma_{X_r}^2 + C_2} \right) \cdot \left(\frac{\sigma_{X_g X_r} + C_3}{\sigma_{X_g}\sigma_{X_r} + C_3} \right), \quad (7)$$

μ_{X_g} and μ_{X_r} represent the mean values of the generated image and the ground truth, σ_{X_g} and σ_{X_r} represent the variance of the two images, $\sigma_{X_g X_r}$ is the covariance, and $C_i, i = 1, 2, 3$, is a constant.

Peak Signal-to-Noise Ratio (PSNR) is a widely used metric for assessing image quality. It is calculated as the ratio of the peak signal energy to the mean square error between the generated frame and the ground truth, providing a measure of the fidelity of the generated image. PSNR is computed as follows:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right), \quad (8)$$

MAX refers to the maximum possible pixel value of the image. For common 8-bit images, the maximum pixel value is 255. *MSE* refers to the mean squared error.

Frechet Inception Distance (FID) [66] evaluates the quality of generated images by measuring the similarity between the distributions of real and generated images in the feature space. This metric calculates the distance between these two distributions, with features extracted from the Inception v3 network [162], to provide an accurate reflection of the image quality. FID is defined by the equation:

$$\text{FID} = \|\mu_g - \mu_r\|_2^2 + \text{Tr} \left(\Sigma_g + \Sigma_r - 2(\Sigma_g \Sigma_r)^{\frac{1}{2}} \right), \quad (9)$$

μ_g and μ_r are the mean values of the feature vectors of the generated image and the ground truth, Σ_g and Σ_r are the covariance matrices and covariance of the feature vectors, and *Tr* is the trace of the matrix.

Learned Perceptual Image Patch Similarity (LPIPS) [221] employs a pre-trained convolutional neural network to extract features from image pairs, weights these features, and then computes the L_2 norm of the weighted features to derive an LPIPS score. It is crucial for assessing image quality, as it correlates closely with human visual perception. For example, using the VGG network as a pre-trained model to calculate:

$$\text{LPIPS} = L_2 \left(\text{VGG}(X_g, X_r) \right), \quad (10)$$

X_g and X_r means the generated image and the ground truth.

Inception Score (IS) [147] quantifies both the quality and diversity of generated images. Despite its utility, IS has limitations and is often used in conjunction with FID for a more comprehensive evaluation. Its formulas are as follows:

$$\text{IS}(G) = \exp \left(E_{\mathbf{x} \sim p_d} D_{KL}(p(y | \mathbf{x}) \| p(y)) \right). \quad (11)$$

Among them, $p(y | x)$ is the conditional probability distribution, where $p(y)$ represents the label set, and x represents the image. $p(y)$ denotes the marginal probability distribution, and $G(z)$ is the image generated by the generator model when provided with a latent vector of random numbers. D_{KL} represents the KL divergence between $p(y | x)$ and $p(y)$.

Landmark Distance (LMD) [24] measures the precision of lip movements in synthesized videos by calculating the Euclidean distance between corresponding landmarks in the lip region, normalized by temporal length and the number of landmarks. The calculation formula is:

$$LMD = \frac{1}{N} \sum_{i=1}^N \sqrt{(x_{g_i} - x_{r_i})^2 + (y_{g_i} - y_{r_i})^2}, \quad (12)$$

N represents the number of feature points. x_{g_i} and y_{g_i} are the x and y coordinates of the i th feature point on the generated image, respectively. x_{r_i} and y_{r_i} are the x and y coordinates of the i th feature point on the real image, respectively.

Average Keypoint Distance (AKD) measures the average distance between key points in the generated image and the corresponding key points in the ground truth. It is used to assess how well the motion from the input driving image is preserved in the generated output. Taking a face alignment technique as an example, which can accurately identify and mark 68 structural coordinate points of the human face, the calculation formula is as follows:

$$AKD = \frac{\sum_{j=1}^{68} |l_j^g - l_j^r|}{68}, \quad (13)$$

l_j^g and l_j^r represent the facial keypoints detected from the generated image and the real image, respectively.

Cosine Similarity (CSIM) is employed to assess the similarity between two N -dimensional vectors by computing the cosine of the angle between them. The CSIM value ranges from -1 to 1, where values closer to 1 indicate higher similarity, and values closer to -1 indicate greater dissimilarity. In this study, CSIM is utilized to evaluate the identity preservation of the generated images. Specifically, we first extract identity-representative embedding vectors of both the source and generated images using the facenet-pytorch library, and then compute the cosine similarity between the two embeddings. The calculation is given by:

$$CSIM = \frac{\mathbf{I}_g \cdot \mathbf{I}_r}{\|\mathbf{I}_g\| \times \|\mathbf{I}_r\|} = \frac{\sum_{i=1}^N (I_g^i \times I_r^i)}{\sqrt{\sum_{i=1}^N (I_g^i)^2} \times \sqrt{\sum_{i=1}^N (I_r^i)^2}}, \quad (14)$$

I_g and I_r represent the identity embedding extracted from the generated image and the real image.

7.3 Experimental Analysis

We compile quantitative results of some classic and SOTA methods on commonly used datasets for both video-driven task and audio-driven task as shown in Table 7 and Table 8. To display the potential optimal results of each paper and to conduct a fair comparison, the data for both tables are taken from the original publications. We select commonly used metrics for comparison, moreover, we invest the different method's performance in the same dataset in order to provide an intuitive reference. Voxceleb [7] is one of the most widely used datasets for video-driven facial animation tasks, and has become a standard benchmark for evaluating and comparing different methods. Among the competing approaches, MCNet [69] demonstrates superior performance across multiple metrics. Notably, it achieves the highest scores in SSIM (0.825) and AKD (1.203) on this dataset, indicating strong capabilities in preserving structural consistency and accurately transferring motion. In addition, it also attains the lowest LPIPS (0.151) among all compared methods, suggesting that it also excels in perceptual quality by generating more realistic and visually consistent facial images. For audio-driven talking head generation, MEAD [178] has emerged as a increasingly adopted benchmark in recent years due to its high-resolution facial data and rich emotional expressions. Within the pool of evaluated methods, DiNET [225] demonstrates

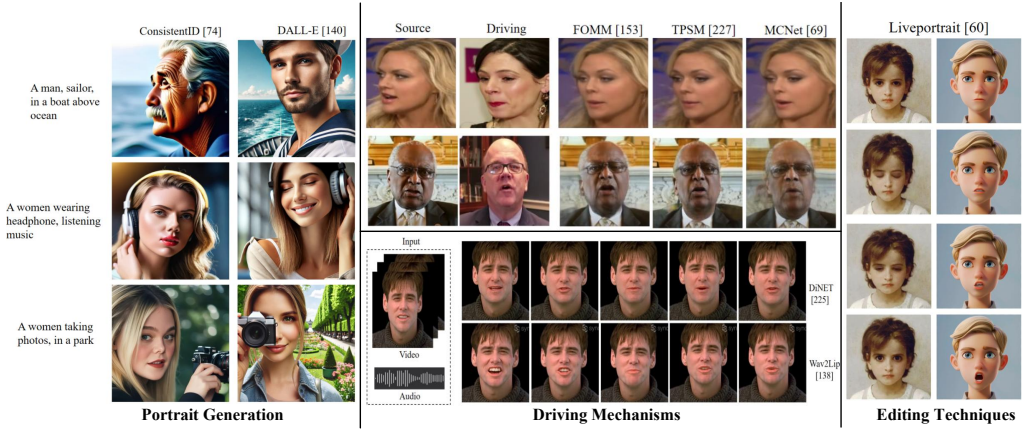


Fig. 5. Examples of common methods, respectively, in portrait generation, driving mechanisms and editing techniques.

more outstanding performance. It achieves the highest SSIM (0.920), indicating superior capability in preserving facial structure. More importantly, DiNET [225] attains the best SyncNet score (7.260) reflecting its exceptional audio-visual synchronization, it also performs competitively on other metrics such as LMD (2.695) and CPBD (0.070) which suggests that it maintains high temporal and perceptual quality. These results highlight it as a state-of-the-art solution for high-fidelity and temporally coherent audio-driven face generation on MEAD [178].

To gain a deeper understanding of the performance and characteristics of different techniques across the three stages of talking head synthesis, we present representative experimental results in Figure 5, covering portrait generation, driving mechanisms, and editing techniques. In the portrait generation section, we compare the results generated by ConsistentID [74] and DALL-E [140] under the same textual descriptions. ConsistentID [74] generates portrait content by inputting a photo while ensuring identity consistency, whereas DALL-E [140] generates high-fidelity portraits solely based on textual prompts. Both methods demonstrate strong capabilities in generating realistic facial images, with ConsistentID [74] excelling in identity consistency and DALL-E [74] outperforming in image details and realism. In terms of driving mechanisms, the visualizations are divided into video-driven and audio-driven methods. The video-driven methods, including FOMM [153], TPSM [227], and MCNet [69], transfer pose and expression information from a driving video to a source image, aiming to maintain identity consistency and accurately animate facial dynamics. MCNet [69] significantly improves temporal coherence and natural motion compared to FOMM [153]. The audio-driven methods, represented by DiNET [225] and Wav2Lip [138], animate facial regions based on speech audio, highlighting the potential of audio-driven techniques in facial expression synthesis. Lastly, LivePortrait [60] demonstrates fine-grained control over facial attributes. By adjusting parameters such as eye closure and mouth movement, it transforms static portraits into dynamic facial sequences, enabling customized expressions and gaze directions without the need for retraining the model, showcasing its flexibility and efficiency in facial animation editing.

8 Applications

Talking head synthesis technology has a wide range of applications across various sectors, revolutionizing how we interact with digital content. This section explores some of the most promising application areas where talking head synthesis is making significant impacts.

Film and Television. Talking head synthesis is transforming visual storytelling in the film and television industry. By employing digital characters, filmmakers can transcend traditional filming

Table 7. Comparison of State-of-the-art Methods in Talking Head Synthesis which Takes an Image and a Video as Inputs

Dataset	Method	Editing	Metrics					
			SSIM↑	PSNR↑	LPIPS↓	FID↓	AKD↓	CSIM↑
Voxceleb [7]	FOMM [153]		0.723	30.390	0.199	-	1.294	0.64
	Face-vid2vid [183]		0.761	30.690	0.212	-	1.620	-
	Bi-layer [215]		-	20.190	0.152	92.200	-	0.65
	DAGAN [70]		0.804	31.220	0.185	-	1.279	-
	MCNet [69]		0.825	31.94	0.174	-	1.203	-
	MarioNETte [216]		0.744	23.240	-	-	-	0.755
	HeadGAN [42]	✓	-	21.460	0.112	36.100	-	0.807
	Free-HeadGAN [41]		-	22.160	0.100	35.400	-	0.81
	HyperReenact [16]		-	-	0.230	27.100	-	0.71
Voxceleb2 [33]	X2Face [191]		0.719	22.540	-	-	7.687	0.6
	FOMM [153]		0.770	23.250	0.090	73.710	2.140	0.74
	Face-vid2vid [183]		0.800	24.370	-	69.130	2.070	0.808
	Bi-layer [215]		0.660	16.980	0.080	203.360	5.380	0.653
	PIRender [142]	✓	-	-	0.264	14.407	-	-
	Metaportrait [217]		-	-	0.226	11.953	-	-
	X2Face [191]		-	-	0.681	45.291	-	-
	ROME [89]		0.860	26.200	0.080	-	-	-
	LIA [187]		-	22.290	-	30.230	1.049	-
	DAGAN [70]		-	25.640	-	24.920	0.844	-
	PECHHead [55]	✓	-	26.960	-	23.050	0.626	-
HDTF [226]	Metaportrait [217]		-	-	0.208	21.497	-	-
	ROME [89]		0.838	20.750	0.173	31.550	2.938	0.746
	OTAvatar [117]	✓	0.806	20.120	0.198	36.630	2.933	0.619
	GONHA [100]		0.868	22.150	0.117	21.600	2.596	0.789
TalkHead-1KH [184]	FOMM [153]		0.790	23.280	0.160	33.220	2.905	0.8805
	Face-vid2vid [183]		0.810	23.590	0.160	35.120	3.100	0.8774
	MRAA [154]		-	25.500	-	32.570	1.057	-
	TPSM [227]		-	25.530	-	32.770	0.983	0.8848
	LIA [187]		-	24.430	-	38.890	0.932	-
	DAGAN [70]		-	23.930	-	34.350	2.405	0.8798
	PECHHead [55]		-	26.760	-	30.100	0.746	-
	IWA [119]		-	23.320	0.150	-	3.480	-

Quantitative evaluations are conducted on various datasets, with the best performance for each metric highlighted in blue.

constraints. This technology enables the seamless creation of intricate close-up shots and high-risk action sequences that are challenging or unsafe to capture with live actors. Additionally, it reduces reliance on expensive special effects in post-production, thereby lowering production costs and accelerating the creative process. The integration of talking head synthesis not only enhances the visual appeal and narrative depth of productions but also opens up new possibilities for innovative storytelling techniques.

Virtual Anchoring. Virtual anchors, as distributors of news and entertainment content, can broadcast continuously, providing uniform performances in both live and pre-recorded formats. They overcome language and cultural barriers by presenting content in multiple languages to cater to global media consumption needs. The flexibility of virtual anchors in style and appearance allows them to quickly adapt to changing market trends and audience preferences.

Game Industry. Talking head synthesis technology significantly enhances the gaming experience by enabling characters to exhibit a wider range of expressions and reactions, thereby increasing player immersion. It allows developers to create characters with rich emotional depth and diverse interactions, offering a unique and interactive gaming experience. As the technology advances, game characters can increasingly reflect players' choices and emotions in real-time, creating personalized gaming.

Table 8. Comparative Overview of Advanced Methods in Talking Head Synthesis Using Image and Audio Inputs

Dataset	Method	Editing	Metrics				
			SSIM↑	PSNR↑	LMD↓	CPBD↑	SyncNet↑
LRW [34]	Wav2Lip [138]		0.862	30.630	5.250	0.152	6.900
	MakeltTalk [231]		0.796	30.380	7.130	0.161	3.100
	ATVG [26]		0.810	30.910	1.370	0.189	-
	PC-AVS [230]	✓	0.778	30.390	3.930	0.185	6.400
	LPGG [25]		0.730	29.650	1.730	-	-
	AVCT [181]		0.805	-	3.560	0.181	-
	EAMM [80]		0.740	30.920	1.610	-	5.520
	FONT [104]		0.825	-	3.480	0.187	-
CREMA-D [20]	You said that citechung2017you		0.770	29.910	1.630	-	-
	Wav2Lip [138]		0.886	34.230	3.320	0.253	5.890
	Audio2Head [180]		0.59	29.760	3.280	-	2.190
	MakeltTalk [231]		0.750	31.370	3.360	0.152	3.500
	PC-AVS [230]	✓	0.610	28.470	3.110	0.127	6.120
	ACDM [12]		0.920	32.470	-	0.290	4.980
	EAMM [80]		0.740	29.430	3.120	0.100	3.980
	RS-DFA [172]		0.844	27.980	-	0.277	-
HDTF [226]	EMMN [164]		0.680	30.030	3.030	-	2.410
	DiNET [225]		0.943	30.008	-	-	6.840
	PC-AVS [230]	✓	0.6383	20.630	3.180	0.164	0.420
	Audio2Head [180]		0.735	-	4.830	0.145	-
	Wav2Lip [138]		0.908	29.288	2.890	0.176	0.630
	MakeltTalk [231]		0.751	19.860	5.460	0.132	0.570
	AVCT [181]		0.769	-	2.710	0.167	0.760
	StyleHEAT [212]		0.805	19.180	-	-	-
MEAD [178]	DreamTalk [116]		0.850	-	2.150	0.310	5.170
	SadTalker [222]		0.770	-	4.070	0.240	4.350
	DiNET [225]		0.920	29.118	-	-	7.260
	PC-AVS [230]	✓	0.775	23.695	4.970	0.070	2.180
	Wav2Lip [138]		0.899	28.539	4.050	0.180	5.260
	GC-AVT [102]	✓	0.340	-	7.100	0.140	2.420
	EAMM [80]		0.400	29.230	6.480	0.080	1.410
	Styletalk [115]	✓	0.840	-	3.250	0.160	3.470
	EMMN [164]		0.660	29.380	2.780	-	3.570
	SadTalker [222]		0.690	-	4.370	0.160	2.760
	PD-FGC [175]		0.490	-	4.100	0.050	2.270
	DreamTalk [116]		0.860	-	2.910	0.160	3.780
	Audio2Head [180]		0.600	29.070	3.220	-	3.200

Quantitative assessments have been performed across multiple datasets with the superior results marked in blue.

Virtual Instructors. Talking head synthesis technology enables the creation of virtual educators capable of real-time interaction and expressive facial responses. These virtual instructors deliver standardized educational content and provide personalized guidance based on students’ reactions and progress. This interactive and individualized approach enhances student engagement and learning efficiency, making it particularly beneficial for remote education and self-study contexts. The use of virtual instructors significantly expands the reach and accessibility of educational resources.

9 Summary and Future Work

In this section, we provide a comprehensive overview of the advantages and limitations of current techniques, as well as potential directions for future research. By examining the strengths and weaknesses of existing methods, we can better understand their impact and identify areas for improvement. Furthermore, we will explore promising avenues for future work that can address current challenges and push the boundaries of what talking head synthesis can achieve.

9.1 Advantage and Limitation

Portrait generation has advanced significantly through various generative methods, both unconditional and conditional. Unconditional methods, such as noise-based generative models, enable the random creation of portraits without requiring prior labels or data inputs, resulting in diverse and abundant image datasets essential for training many deep learning models. Conditional methods, including text-guided image synthesis, provide controlled and specific outputs based on descriptive inputs, enhancing the relevance of generated portraits for specific applications. Despite these advancements, achieving consistently high-quality and realistic portraits remains challenging. Due to complex geometry and appearance, existing methods face issues like inflexibility, instability, and low fidelity. While large models like diffusion models offer promise, they demand substantial computational resources.

The driving mechanism for animating still facial images, or talking head synthesis, has shown impressive performance with both audio-driven and video-driven methods. Audio-driven methods use vocal or text inputs to animate facial features synchronously, while video-driven methods replicate facial expressions and motions from a driving video. Advanced deep learning techniques, including GANs and VAEs, have significantly improved the real-time rendering capabilities of animated portraits, making them viable for interactive applications such as virtual meetings or customer service avatars. However, persistent challenges such as temporal inconsistency and poor results when there are extreme differences between the source image and the driving image hinder further improvement in generation quality. Additionally, the complexity of current models necessitates deep technical knowledge and extensive training data to achieve optimal results.

Editing in talking head synthesis, whether 2D-based or 3D-based, has evolved to offer significant control over the modification of generated avatars. Techniques for disentanglement and decoupling allow for the independent manipulation of specific facial features without affecting the overall structure, enhancing flexibility in post-generation customization. Direct parametric control and multimodal integration enable comprehensive adjustments that are coherent across different modalities. However, decoupling itself is challenging due to the interconnected nature of various attributes, and the effectiveness of decoupling directly influences the quality of the results. Ensuring harmony and consistency among attributes is crucial. While 2D-based methods are limited in fine control due to the lack of geometric information, 3D-based methods excel in this aspect but require extensive training time and computational resources.

Recent advancements in portrait generation, driving mechanisms, and editing techniques have been significant, yet persistent challenges remain. The integration and complementarity among these domains have opened new possibilities for creating dynamic and lifelike digital portraits but also highlight areas needing further research and optimization. Section 9.2 addresses pivotal issues and provides insights into potential future directions for tackling these challenges.

9.2 Future Work

Building on the limitations identified above, we outline prospective research directions that we categorize into four inter-related dimensions: *model-level challenges*, *data and generalization issues*, *system-level integration*, and *ethical as well as societal considerations*. This thematic grouping clarifies current bottlenecks and highlights the most promising paths toward next-generation talking-head systems.

9.2.1 Model-Level Challenges.

Dependency on Pre-trained Models. State-of-the-art pipelines frequently inherit large diffusion or autoregressive backbones. Although such inheritance accelerates progress, it inevitably constrains architectural innovation and hampers adaptation to emerging use cases. Future work

should explore training lightweight yet expressive modules on domain-specific corpora, or devise hybrid schemes that combine generic pre-trained priors with task-tailored networks.

One-shot Image Input. A single portrait rarely encodes sufficient geometry and appearance cues for dynamic avatar generation, leading to jerky motion or identity drift under novel views. Advances in ultra-light 3D reconstruction and implicit neural representation could mitigate this limitation by supplying richer priors from minimal input.

Large-Angle Driving. Scarcity of extreme-pose data still degrades synthesis quality when the head turns beyond typical conversational ranges. Enriching multi-view collections and designing pose-robust networks—e.g., via view-consistent 3D supervision—are critical steps forward.

Temporal Consistency and Jitter. Lapses in long-range dependency modeling manifest as temporal flicker and jitter. Promising directions include recurrent or transformer-style temporal modules with explicit physics-inspired constraints, as well as adversarial training tailored to sequence coherence.

Trajectory Consistency in Expression Editing. A distinctive challenge in talking head editing lies in maintaining semantic and temporal coherence of editing trajectories across expression sequences. While most existing approaches apply editing operations frame by frame, they often fail to ensure smooth transitions between intermediate states, resulting in visible motion discontinuities or unnatural expression changes. Future work should explore trajectory-aware editing frameworks that incorporate consistency constraints in latent space transitions or use trajectory prediction modules to regulate the dynamics of facial attribute modifications over time.

9.2.2 Data and Generalization Challenges.

Data Bias. Imbalanced distributions of ethnicity, age, and capture conditions propagate into unfair or stereotyped outputs. Attribute-aware sampling, adversarial de-biasing, and controllable synthesis interfaces can increase both fidelity and inclusiveness.

Multilingual Speech. Most lip-sync systems are optimized for English, whereas languages such as French, German, or Brazilian Portuguese exhibit distinct phonetic patterns [11, 38, 166]. Cross-lingual meta-learning and phoneme-agnostic viseme modeling constitute important research avenues.

Expressionless Lip Synchronization. Purely acoustic control often yields accurate lip shapes yet bland facial affect. Coupling fine-grained emotion or semantic recognition with expression priors may enrich facial expressivity while preserving audio-visual alignment.

Complexity-aware Editing. The difficulty of editing is highly correlated with the structural and semantic complexity of the target portrait. When dealing with images that feature intricate facial geometries, occlusions (hands, glasses, and hair), or highly expressive motions, current editing methods often struggle to produce satisfactory results.

Semantic Accuracy in Interactive Tasks. In interactive scenarios where editing intentions are expressed through vague or subjective descriptions such as “a more confident smile” or “slightly raised eyebrows.” These commands lack clear quantitative definitions, making it difficult for current models to interpret and execute them consistently. Addressing this challenge requires bridging the gap between human-intended semantics and model-controllable parameters.

9.2.3 System-Level Integration and Joint Optimization. **Identity Coherence.** Small discrepancies in identity embedding as it passes from portrait generation through motion driving and editing can accumulate into perceptible drift over long sequences. Future work should explore cross-stage regularizers—such as shared identity codes or cycle-consistent mappings—to enforce a stable identity manifold across heterogeneous representations.

Interface Compatibility. Portrait latents, key-point or mesh geometries, and parametric control vectors use fundamentally different data structures. Without a standardized intermediate

representation or robust translation modules, pipeline components remain brittle and hard to combine. A formally defined, differentiable “lingua franca” is essential to enable plug-and-play experimentation and scalable architecture.

Computational Synchronization. The three stages exhibit markedly different computational costs and latency profiles. Naïvely chaining them leads to hardware underutilisation or unpredictable frame-rate drops, breaking real-time constraints. Promising solutions include latency-aware orchestration, asynchronous pipelining, and lightweight surrogate models that adaptively offload or cache expensive computations.

End-to-end Optimization. Current pipelines are typically trained stage-wise under disjoint losses, which maximises local performance at the expense of global coherence. A holistic approach requires designing fully differentiable architectures with multi-objective loss functions that balance identity fidelity, expression accuracy, temporal smoothness, and inference efficiency. Techniques such as gradient-stable interface layers and curriculum learning may help resolve conflicting objectives.

9.2.4 Ethical and Societal Considerations. Deepfake Risks and IP Compliance. Higher photorealism amplifies the potential for defamation, fraud, or political misinformation, while repurposed likenesses trigger copyright and personality-rights disputes. Robust watermarking, deepfake detection, and provenance tracking—alongside clear governance frameworks—are indispensable safeguards.

Privacy and Informed Consent. Training data often contain identifiable faces collected without explicit permission. Future pipelines must incorporate consent-aware data curation and privacy-preserving learning (e.g., federated or differential approaches).

Transparency and Accountability. End-users should be able to verify whether a video is synthetic and trace its generative lineage. Model cards, usage policies, and auditable release practices can enhance public trust and support responsible deployment.

10 Conclusion

This article has systematically reviewed and analyzed the key areas of talking head synthesis, including portrait generation, driving mechanisms, and editing techniques. Each domain was explored in detail, highlighting significant papers and summarizing their strengths and limitations. We also compiled comprehensive datasets and evaluated existing techniques using a diverse set of metrics. Furthermore, we discussed widespread applications and outlined future research directions.

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